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Representational Measurement of Similarity: The Additive-Difference Model, Revisited

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Contents

Symbols

Throughout this thesis, we follow a consistent notation and try to use conventional notations and notations introduced in the used literature. Here we will provide a quick overview over the notation.

$\mathbb{R}$ and $\mathbb{Q}$ denote the real and rational numbers respectively, with restrictions indicated via subscripts (e.g., $\mathbb{R}_{>0}$ and $\mathbb{R}_{\geq 0}$). Other sets are generally denoted with uppercase letters, except for the letters F and G, which are real-valued functions from the AD model and are used throughout this thesis.
Dimensions are represented by n and i, k and m are integers used for indexing or counting.
Vectors are bold (e.g., $\mathbf{x}, \mathbf{y}$) with coordinates $\mathbf{x} = (x_1, \ldots, x_n)$. Sequences of vectors are indexed as $\mathbf{x_i}$ or $\mathbf{x^{(i)}}$.
Greek letters are either real-valued functions and have a special meaning (such as φ, δ or γ) or constants defined in the context (e.g., α, β). Other symbols are used according to conventions, e.g., λ for convex combinations, ε for small real quantities and $|x|$ for the absolute value of x.

Any symbols not mentioned here will be introduced as needed within the relevant sections.

Metric Spaces and Topology

$\langle X, \delta \rangle$	A metric space
$\mathrm{B}^{\delta}_{(r)}$ / $\mathrm{B}^{\delta}_{[r]}$	The open (r) or closed $[r]$ ball with radius r around 0 w.r.t. the metric δ
δ	A metric function. Depending on the context, this refers to an arbitrary metric or the metric from the segmentally additive AD model
δ_r	A Minkowski (or power) metric
δ_∞	The maximum metric (Tchebychev distance)
γ	A parametric curve
$\mathrm{L}(\gamma)$	The length of a parametric curve
$\mathrm{conv}(A)$	The convex hull of A

extreme(A)	The extreme points of A

Proximity Structures

$\langle \mathcal{A}, \succsim \rangle$	A proximity structure, most of the time a factorial PS. Elements denoted by a, b, c and so on.
$\langle abc \rangle$	The ternary relation on a proximity structure
$a\|b\|c$	a, b and c differ only on one factor, and b is between a and c

Functions

φ_i	A real-valued function defined on a proximity structure
F_i	The coordinate function in the AD Model

Other symbols

$\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$	The quasi-arithmetic ϕ-mean of $\mathbf{x}$ with weights $\mathbf{w}$
$\mathcal{M}_p(\mathbf{x}, \mathbf{w})$	For $p > 0$ the p-th power mean, for $p = 0$, the geometric mean of $\mathbf{x}$ with weights $\mathbf{w}$

Mathematical Prerequisites

This thesis assumes familiarity with fundamental concepts from real analysis, typically covered in introductory courses. These include limits, continuity, differentiation, monotonicity, metric spaces and basic notions of topology. Familiarity with metric spaces is highly beneficial. In Chapter 3, we introduce geometric notions, particularly curves and segments in metric spaces. Furthermore, we require results on convexity, especially in Chapter 4, where we introduce all the necessary concepts. For reference, see any textbook on fundamental analysis, such as Thomson et al. (2008).

Chapter 1

Introduction

In the beginning of the 19th century the physicist and philosopher N. R. Campbell heavily criticized psychology and the social sciences, arguing that, in contrast to physics, it is impossible to construct measurement in these areas (Campbell, 1957). However, these claims were proven to be incorrect. One of the first psychologists that responded to this critique was S. S. Stevens, who argued that Campbell's view on measurement was too narrow (Stevens, 1951, 1975). Around the same time he developed the four scales of measurement presented in Stevens (1946), which are introduced in beginner psychology lectures and are fundamental to his view of measurement (R. D. Luce & Suppes, 2002, p. 14). Later, philosophers, mathematicians and psychologists, namely Patrick Suppes, David H. Krantz, R. D. Luce, Amos Tversky, amongst others, developed the representational measurement theory, which also proved the claims of Campbell to be incorrect. This theory aims to create a more rigorous and axiomatic foundation for measurement and can be found in the three volumes of Foundations of Measurement Krantz et al. (1971, 1989) and D. Luce et al. (1990). The structures developed in these volumes serve as examples that measurement in psychology is possible.

Representational measurement is expressed in mathematical terms [1] and begins with the observation that there are essentially two structures – one empirical and one numerical – which should be connected by a homomorphic function (Krantz et al., 1989, p. 8). We explain what we mean by that with an example.
For example, one could measure the perceived loudness of sounds in an experiment, and ask the participants to rate which of two presented sounds is louder. In this case, the empirical structure is the set of the sounds presented in the experiment, denoted by $\mathscr{A}$. The relation on this set is the ordering of loudness, i.e., whether

[1] For mathematical prerequisites of this thesis see Mathematical Prerequisites.

the loudness of the sound a is greater than or equal to that of sound b, denoted by $a \succsim b$. Together, we denote this as the *empirical structure* $\langle \mathcal{A}, \succsim \rangle$.
One natural choice, is to try to map this structure into the positive real numbers with the ordinary "greater or equal to" relation. Then, $\langle \mathbb{R}_{\geq 0}, \geq \rangle$ is the corresponding *numerical structure*.
This means that we search for a function φ that maps each sound a to a real number $\varphi(a)$, which we can call the "loudness" of sound a. Naturally, we want the function φ to preserve the ordering by $\succsim$. If sound a is perceived louder than b (i.e., $a \succsim b$), we want the corresponding numbers to be in the same order, i.e., $\varphi(a) \geq \varphi(b)$. A function satisfying these properties is called a *homomorphism*.

The goal of representational measurement is to establish conditions under which such a homomorphism exists and to which degree it is unique. A *representation* (or *existence*) theorem establishes the existence of a certain representation given some conditions, and a *uniqueness* theorem answers the question of uniqueness. The conditions that are used in these theorems are called *axioms* and can be seen as empirical laws that may or may not be true for certain applications (Krantz et al., 1989, pp. 6-13). Homomorphisms that map into the real numbers are called scales. We will briefly introduce the four scales of measurement developed by Stevens, ranked by their uniqueness (this corresponds to the amount of admissible transformations that preserve the homomorphism properties).

Definition 1.0.1 (Four Scale Types)**.**

1. *Nominal scale*: Scale where only one-to-one assignments are made. The permissible transformations are all bijective mappings. Examples include genders (e.g., in questionnaires: 1 = female, 2 = male, 3 = diverse) or postal codes.

2. *Ordinal scale*: In addition to one-to-one assignments, there is an order. Therefore, the permissible transformations are strictly monotonic mappings. Examples include grades (1 to 6) or earthquake magnitude.

3. *Interval scale*: In addition to order, the difference between values matter. The permissible transformations are affine linear mappings $f(x) = \alpha x + \beta, \alpha > 0$. Examples include the Celsius and Fahrenheit scales.

4. *Ration scale*: In addition to the differences in the interval scale, there is a fixed zero point. Therefore, the permissible transformations are all linear mappings $f(x) = \alpha x, \alpha > 0$. Examples include length or mass.

If we consider similarity of colors, for example, we cannot use the empirical structures described above as there is no simple $\succsim$ relation for colors i.e., there is no direct way to compare two colors. However, it is possible to compare pairs by asking whether colors a and b are more similar than colors c and d. This gives rise to proximity structures, a type of empirical structure commonly used to infer how the data is structured and which elements are close to each other. In this thesis we focus on proximity structures and present their numerical representations along with the associated representation and uniqueness theorems. We begin by defining proximity structures and the notation used above. At the end of the next section, we provide a brief overview of the chapter's content.

1.1 Proximity Structures and Metrics

To be able to compare two stimuli pairs in a mathematical sense, we introduce a relation on a stimuli set that compares two stimuli pairs. If we have four stimuli a, b, c and d, then $(a, b) \succsim (c, d)$ denotes that the stimuli a and b are equal or less similar to each other than the stimuli c and d (or more precisely, the dissimilarity between a and b is at least as great as that between c and d). Analogously, $\sim$ means that the similarity of two pairs is equal. The essence of a *proximity structure* is precisely that, with some additional constraints.

Definition 1.1.1 (Proximity Structure)**.**
Suppose $\mathcal{A}$ is a set and $\succsim$ a quaternary relation on $\mathcal{A}$, i.e., a binary relation on $\mathcal{A} \times \mathcal{A}$. Then $\langle \mathcal{A}, \succsim \rangle$ is a *proximity structure* (PS) iff (if and only if) the following axioms hold for all $a, b \in \mathcal{A}$:

1. $\succsim$ is a weak ordering on $\mathcal{A} \times \mathcal{A}$, i.e., it is connected and transitive.
2. $(a, b) \succsim (a, a)$ whenever $a \neq b$.
3. $(a, a) \sim (b, b)$ (minimality).
4. $(a, b) \sim (b, a)$ (symmetry).

The structure is *factorial* iff

$$\mathcal{A} = \prod_{i=1}^{n} \mathcal{A}_i.$$

Remark (Connectedness). Connected means, that comparison of all distinct elements is possible, i.e., for a relation R on a set X it holds for all $x, y \in X$ with $x \neq y$ that either xRy or yRx.

We briefly explain these axioms. The first axiom is the ordering assumption, which serves as the basis for proximity measurement. Axiom 2 imposes a natural constraint: other-dissimilarity must exceed self-dissimilarity. Axiom 3 requires self-dissimilarity to be equivalent across all objects, and axiom 4 require dissimilarity to be symmetric. In the final chapter, we will discuss these axioms – along with those introduced in the following chapter – in more detail, particularly regarding their empirical validity. The mathematical concept we aim to connect proximity structures to is the concept of a metric. A metric is defined as follows:

Definition 1.1.2 (Metric)**.**
Let X be a set. A real-valued function δ on $X \times X$ is a *metric*, iff the following properties hold for all $x, y, z \in X$:

1. $\delta(x, x) = 0$ and $\delta(x, y) > 0$ if $x \neq y$

2. $\delta(x, y) = \delta(y, x)$

3. $\delta(x, y) + \delta(y, z) \geq \delta(x, z)$

We call $\langle X, \delta \rangle$ a *metric space.*

Metrics are measures of distance in a space. The third property is called the *triangle inequality* and is central to the intuition of measuring distances. The triangle inequality can be formulated as follows: Adding a detour through y on the way from x to z can only increase the distance.

Note, that $\langle \mathbb{R}_{\geq 0}, \geq \rangle$, the image of a metric, is the desired numerical structure that we aim to connect to a proximity structure (i.e., the empirical structure). Therefore, δ is our desired homomorphism in the sense that it should transform $\succsim$ into $\geq$. If such a metric δ exists, then we say that the proximity structure is representable by a metric. This is captured in the next definition.

Definition 1.1.3 (Representability of $\succsim$ by a Metric)**.**
A proximity structure $\langle \mathcal{A}, \succsim \rangle$ is *representable by a metric* iff there exists a real-valued function δ on $\mathcal{A}$ such that:

1. $\langle \mathcal{A}, \delta \rangle$ is a metric space, i.e., δ is a metric on $\mathcal{A}$.

2. δ preserves the relation $\succsim$, i.e., for all $a, b, c, d \in \mathcal{A}$

$$\delta(a, b) \geq \delta(c, d) \text{ iff } (a, b) \succsim (c, d).$$

Krantz et al. (1971, p. 162) prove that representability by a metric does not impose any empirical constraints. However, simple metrics do not provide enough structure, and we therefore require the metrics to satisfy additional conditions, which can be observed in often used metrics. One of the most used metrics is the Minkowski p-metric or Minkowski distance (also called the power or L^p metric) and is defined in the following way.

Definition 1.1.4 (Minkowski Metric)**.**
For $n \in \mathbb{N}_{>0}$ and $p \geq 1$ the *Minkowski Metric* of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as

$$\delta_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

For $p \geq 1$, this is indeed a metric, as the Minkowski inequality establishes the triangle inequality (Bullen, 2003, pp. 189-191 Theorem 9).

If we closely inspect this metric, we can observe two interesting properties. The first one, arises from the way the metric is computed, which can be broken down into four steps.

1. Decompose two vectors $\mathbf{x}$ and $\mathbf{y}$ into their underlying coordinates $(x_1, \ldots x_n)$ and $(y_1, \ldots, y_n)$.

2. Calculate the coordinatewise differences $|x_i - y_i|$ and transform them with a bijection $F(x)$ (in this case x^p).

3. Sum up the transformed differences for all coordinates.

4. Transform the sum using the inverse F^{-1} (in this case $x^{1/p}$).

This generalization of the metric yields the *Additive-Difference Model* (Eq. (2.4)) which will be covered in Chapter 2.

The second property that can be observed in the Minkowski Metric is the fact that distances across straight lines are additive (we will show this in Example 3.3.1). This is not the case for all metrics, but only for a special class, called *metric with additive segments*, which will be studied in Chapter 3.

In Chapter 2 and Chapter 3, we will inspect the constraints that each representation imposes on the proximity structure and deduce necessary axioms. These axioms then lead to new structures which allow such representations. This is shown by the representation and uniqueness theorems which are presented at the end of both chapters. We do not prove these theorems, for that refer to Krantz et al. (1971, Chapter 14). Here, we will focus on combining both representations. This leads to

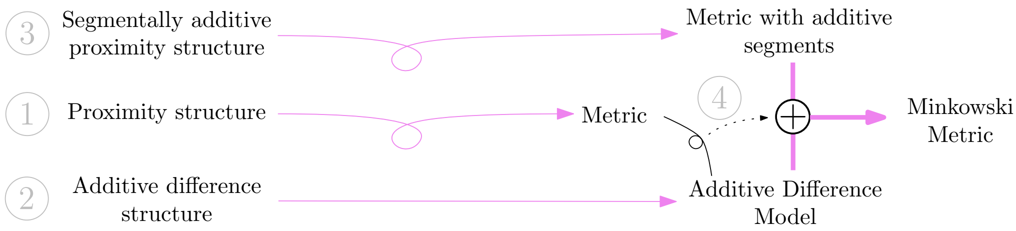

Figure 1.1: General overview of the content, which can be revisited while reading the chapters.
Purple lines represent theorems and gray numbers are the associated chapter numbers. Lines with a circle denote that the given structure is representable by a metric. Our focus is on chapter 4. First, we will combine the AD model with a metric and use these results to, secondly, combine it with a metric with additive segments to obtain the Minkowski metric. The dotted black line indicates that the results from combining a metric with the AD model are used in the main theorem.

the main result of this thesis: the Unique Representation Theorem for Segmentally Additive AD Models (Theorem 4.2.1), which states that the Minkowski metric is the only additive difference metric that is also segmentally additive. Most of Chapter 4 is devoted to proving this theorem. For a visual representation of the content, see Figure 1.1. In the last chapter we discuss the usage of proximity structures and the axioms together with their empirical testability.

Chapter 2

Additive Difference Models and Structures

In this chapter we will establish the additive difference model and additive difference structures which lead to representation with the AD model. At the end of the chapter we will provide an example for an AD model that is a metric. This chapter is based on the first four subsections of Krantz et al. (1971, Chapter 14.4, pp. 175-185). A more elaborate explanation of the references can be found at the end of this chapter.

2.1 Additive Difference Model

As mentioned in Section 1.1 we wish to achieve a generalization of the Minkowski metrics by inspecting 3 general properties that it has. Namely, these are decomposability, intradimensional subtractivity and interdimensional subtractivity which correspond to the first three steps mentioned after Definition 1.1.4 in the discussion of the Minkowski metric. Here, we will define these terms for proximity structures and then remark how they translate to the reals.

Definition 2.1.1 (Additive-Difference Model).
Given a factorial proximity structure $\langle \mathcal{A}, \succsim \rangle$ and two stimuli $a, b \in \mathcal{A}$. A numerical representation of the proximity structures satisfies

1. *Decomposability*, if the distance between stimuli is a function of componentwise contributions. Then

$$\delta(a, b) = F[\psi_1(a_1, b_1), \ldots, \psi_n(a_n, b_n)], \tag{2.1}$$

where F is an increasing [1] real-valued function in each of its n real arguments and ψ_i is a real-valued symmetric function satisfying $\psi_i(a_i, b_i) > \psi_i(a_i, a_i)$ when $a_i \neq b_i$ for all $i = 1, ..., n$.

2. *Intradimensional subtractivity*, if each componentwise contribution is the absolute value of an appropriate scale difference. Then

$$\delta(a, b) = F[|\varphi_1(a_1) - \varphi_1(b_1)|, \ldots, |\varphi_n(a_n) - \varphi_n(b_n)|], \tag{2.2}$$

where F is as in Equation (2.1) and ψ_i is replaced with $|\varphi_i(a_i) - \varphi_i(b_i)|$ for some real-valued function φ_i defined on $\mathscr{A}_i$ for $i = 1, ..., n$.

3. *Interdimensional additivity*, if the distance is a function of the sum of componentwise contributions. Then

$$\delta(a, b) = G\left[\sum_{i=1}^{n} \psi_i(a_i, b_i)\right], \tag{2.3}$$

where ψ_i are as in Equation (2.1) and G is an increasing function in one argument.

4. The *Additive-difference model* (AD model), if both, additivity and subtractivity are assumed. Then

$$\delta(a, b) = G\left\{\sum_{i=1}^{n} F_i\left[|\varphi_i(a_i) - \varphi_i(b_i)|\right]\right\}, \tag{2.4}$$

where G and F_i, for $i = 1, \ldots, n$, are all increasing functions in one argument.

Remark 2.1.2. As noted above, this definition is formulated for proximity structures. However, we wish to make use of the model without having to rely on a proximity structure as the domain. For example the Minkowski metric clearly is a case of the AD model. Thus, when talking about the AD model in the context of the real numbers, we interpret the functions that operate on the proximity structure, i.e., ψ and ϕ, as arbitrary real numbers, since these functions return real numbers.

2.2 Additive Difference Structure

In this section we will develop necessary and sufficient conditions for a proximity structure to be representable by an AD Model. We do this by examining each of the

[1] We use the terms increasing and decreasing to denote real-valued functions that are strictly monotonically increasing and decreasing, respectively.

characteristics of the AD model, namely decomposability, intradimensional subtractivity and interdimensional additivity, and deriving conditions for a representation. All of these conditions are closely related to the theory developed in Krantz et al. (1989) and most of the theorems utilize some sort of reduction to the structures developed there. At the end of this subsection, we will state the representation and uniqueness theorem. As decomposability is the basis assumption of each of the variants, we begin with it.

Decomposability implies that we can represent the observed proximity of two stimuli pairs, characterized by a factorial representation, by a function that evaluates the proximity on each factor. Using the monotonicity in each argument, its necessary that each factor contributes its effect to the overall similarity independently of all the other factors. In particular, for two stimuli pairs that only differ on the i-th dimension, the ordering must be preserved, when the values on the remaining factors are changed in the same way. This is stated in the next definition, where (a, b) and (a', b') are the original stimuli and c, d, c' and d' are the stimuli we obtain by "changing" the remaining dimensions, as just described.

Axiom 2.2.1 (One-Factor Independence)**.**
A factorial proximity structure $\langle A, \succsim\rangle$ satisfies *one-factor independence* iff the following holds for all $a, a', b, b', c, c', d, d' \in A$: If the two elements in each of the pairs (a, a'), (b, b'), (c, c'), (d, d') have identical components on all but one factor, and the two elements in each of the pairs (a, c), (a', c'), (b, d), (b', d') have identical components on the remaining factor, then

$$(a, b) \succsim (a', b') \quad \text{iff} \quad (c, d) \succsim (c', d').$$

This notion of one-factor independence induces an ordering on each factor. Therefore, one-factor independence asserts that the induced ordering is well-defined, i.e., independent of the fixed components on the remaining factors. This induced ordering is denoted by $\succsim_i$. This ordering is of importance for developing intradimensional subtractivity, which we do next.

Intradimensional subtractivity is obtained by establishing decomposability for the entire structure and then imposing an absolute-difference structure from Krantz et al. (1989, pp. 170-173, Definition 4.8) on each factor $\langle \mathscr{A}_i^2, \succsim_i\rangle$. Therefore, the axioms for intradimensional subtractivity are essentially the same as for absolute-difference structures, but reformulated for proximity structures. In absolute-difference structures, just as in proximity structures, the basis is an ordering of pairs which is exactly $\langle \mathscr{A}_i^2, \succsim_i\rangle$ for each $i = 1, \ldots, n$.

Since absolute-difference structures are unidimensional, the comparison of two pairs can be seen as the comparison of two intervals, and therefore the ordering can be seen as an ordering of intervals. Further, the axioms for absolute-difference structures are also unidimensional. These axioms are based on the notion of unidimensional betweenness which is very intuitive, i.e., betweenness of real numbers, but formulated in terms of the ordering relation. The same holds for the betweenness axioms. Since our representation is in the reals, these intuitive concepts must apply to the empirical structure, and therefore are necessary conditions. Next, we present the definitions of betweenness and the betweenness axioms. Note that these can be easily verified for the reals using 6 points on a line.

Axiom 2.2.2 (Betweenness and Betweenness Axioms).
Let $\langle A = \prod_{i=1}^{n} A_i, \succsim \rangle$ be a factorial proximity structure that satisfies one-factor independence. Let $\succsim_i$ denote the induced order on A_i^2. We say that b is *between* a and c, denoted $a|b|c$, if and only if

$$(a_i, c_i) \succsim_i (a_i, b_i), (b_i, c_i) \quad \text{for each } i.$$

We say that $\langle A, \succsim \rangle$ satisfies the *betweenness axioms* if and only if the following hold for all $a, b, c, d, a', b', c' \in A$:

1. Suppose a, b, c, d differ on at most one factor, and $b \neq c$, then
   (i) if $a|b|c$ and $b|c|d$, then $a|b|d$ and $a|c|d$;
   (ii) if $a|b|c$ and $a|c|d$, then $a|b|d$ and $b|c|d$.
2. Suppose a, b, c, a', b', c' differ on at most one factor, $a|b|c$, $a'|b'|c'$, and $(b, c) \sim (b', c')$, then
$$(a, c) \succsim (a', c') \iff (a, b) \succsim (a', b').$$

The betweenness axioms are, however, not enough to establish an absolute-difference structure on each factor. We further need two technical axioms, which are not empirically testable. The first one is restricted solvability, which is very similar to segmental solvability but assures the existence of an element with respect to unidimensional betweenness, i.e., $a|b|c$ instead of $\langle a, b, c \rangle$. Given three stimuli d, a and c we measure the dissimilarities with respect to d and observe the pairs (d, a) and (d, c). Then, for every possible dissimilarity between (d, a) and (d, c), there exists an element b which realizes this dissimilarity.

Axiom 2.2.3 (Restricted Solvability).
A factorial proximity structure $\langle A, \succsim \rangle$ satisfies *restricted solvability* if and only if, for all $e, f, d, a, c \in A$, if $(d, a) \succsim (e, f) \succsim (d, c)$, then there exists $b \in A$ such that $a|b|c$ and $(d, b) \sim (e, f)$.

The second axiom is the Archimedean axiom. The statement of this axiom can be formulated as follows: Given a distance, a radius, and a center, every sequence of points starting from the center and moving away from the center with steps greater than the given distance exits the given radius in a finite amount of steps. In other words, either the steps get smaller or the sequence moves out of the radius. This can be formulated with respect to $\succsim$ as follows.

Axiom 2.2.4 (Archimedean axiom).
A proximity structure $\langle A, \succsim \rangle$ satisfies the *Archimedean axiom* if and only if, for all $a, b, c, d \in A$ with $a \neq b$, any sequence $e^{(k)} \in A$, $k = 0, 1, \dots$ such that $e^{(0)} = c$, $(e^{(k)}, e^{(k+1)}) \succ (a, b)$, and $(c, d) \succ (c, e^{(k+1)}) \succ (c, e^{(k)})$ for all k is finite.

Given these axioms, we can establish an absolute-difference structure on each factor and therefore get an intradimensionally subtractive representation.

Now, we investigate some consequences of interdimensional additivity. For two stimuli a and a' that coincide on the i-th factor, and let the same hold for b and b'. Then $\psi_i(a_i, b_i) = \psi_i(a'_i, b'_i)$, and since our representation is additive, the relation $(a, b) \succsim (a', b')$ must be independent of the i-th component. Moreover, changing the common value of a and a' and the common value of b and b' should preserve the ordering. Therefore, the ordering is independent of the common value. This independence is different from one-factor independence which we defined above and, in fact, is stronger, as it implies one-factor independence but not conversely.

Axiom 2.2.5 (Independence).
A factorial proximity structure $\langle A, \succsim \rangle$ satisfies *independence* if the following holds for all $a, b, c, d, a', b', c', d' \in A$:

1. If the two elements in each of the pairs $(a, a'), (b, b'), (c, c'), (d, d')$, have identical components on one factor, and

2. the two elements in each of the pairs (a, c), (b, d), (a', c'), (b', d'), have identical elements on the remaining factors, then

$$(c, d) \succsim (c', d') \quad \text{iff} \quad (a, b) \succsim (a', b').$$

Given this, and that every factor is essential, i.e., each factor contributes to the ordering in the sense, that there exist $a, b \in \mathcal{A}$ that coincide on all but the one factor satisfy $(a, b) \succ (a, a)$. Using independence we can eliminate all inessential factors and assume that n denotes the number of essential factors.

Taking together all the axioms just defined yields the AD model as a representation, as the next theorem shows. Before stating the representation and uniqueness theorem, we first define additive difference structures as the structures satisfying the axioms we just derived. Note, that one factor independence is implied by independence and therefore is omitted.

Definition 2.2.6 (Additive-Difference Structure).
A factorial proximity structure $\langle \mathcal{A}, \succsim \rangle$ with $n \geq 3$ Factors is an *additive-difference structure* (AD Structure), iff the following axioms hold:

1. Independence
2. Betweenness axioms
3. Restricted solvability
4. Archimedean axiom

Theorem 2.2.7 (Representation and Uniqueness Theorem for AD Structures).
Let $\langle \mathcal{A}, \succsim \rangle$ be an additive-difference factorial proximity structure. Then there exist real-valued functions $\varphi_1, \ldots, \varphi_n$ defined on $\mathcal{A}_1, \ldots, \mathcal{A}_n$, and increasing functions $F_1, \ldots, F_n$ each defined on $\mathbb{R}$, such that for all $a, b, c, d \in \mathcal{A}$,

$$(a_1, \ldots, a_n, b_1, \ldots, b_n) \succsim (c_1, \ldots, c_n, d_1, \ldots, d_n)$$
$$\text{iff}$$
$$\sum_{i=1}^{n} F_i(|\varphi_i(a_i) - \varphi_i(b_i)|) \geq \sum_{i=1}^{n} F_i(|\varphi_i(c_i) - \varphi_i(d_i)|).$$

Furthermore, the φ_i are interval scales, and the F_i are interval scales with a common unit ($i = 1, \ldots, n$).

Remark 2.2.8. For $n = 2$ another axiom, the Thomsen condition, must be assumed. For $n \geq 3$ this conditions follows from the other axioms, see Krantz et al. (1971, pp. 182-184).

2.3 Examples

Example 2.3.1 (p-exp Metric Family)**.**
Let the F in the AD model (Eq. (2.4)) be given by

$$F(x) = \alpha \cdot (e^{q \cdot x} - 1)^p \qquad \alpha, q > 0, p \geq 1. \tag{2.5}$$

The model defined by this F is a metric.

Proof. First, we will prove positivity and symmetry of this function, and then turn to the triangle inequality which is a result of theorem which provides a generalization of the Minkowski inequality.

Part 1 (Positivity and Symmetry)**.**
Positivity and symmetry are both straight forward.

1. Positivity: $\delta(\mathbf{x}, \mathbf{y}) = 0$ implies $F(|x_i - y_i|) = 0$ for all $i = 1, \ldots, n$, which is only possible if $|x_i - y_i| \cdot r = 0$, meaning $x_i = y_i$ for all $i = 1, \ldots, n$. Conversely, if $\mathbf{x} \neq \mathbf{y}$, then $x_i \neq y_i$ for at least one i in $1, \ldots n$, which implies $F_i(|x_i - y_i|) > 0$, and thus $\delta(\mathbf{x}, \mathbf{y}) > 0$.

2. Symmetry: Symmetry follows directly from the fact that $|x_i - y_i| = |y_i - x_i|$ holds.

The triangle inequality follows from Mulholland's Inequality (Mulholland, 1949, p. 297).

Theorem 2.3.2 (Mulholland's Inequality)**.**
The inequality

$$f^{-1}\left[\sum_{i=1}^{n} f(|x_i + y_i|)\right] \leq f^{-1}\left[\sum_{i=1}^{n} f(|x_i|)\right] + f^{-1}\left[\sum_{i=1}^{n} f(|y_i|)\right]$$

holds for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, n = 1, 2, \ldots$ *whenever the following three conditions are satisfied:*

1. f *is continuous, increasing and* $f(0) = 0$.

2. f *is convex on* $[0, \infty)$.

3. $\log f(e^x)$ *is convex for* $x \in (-\infty, \infty)$.

Part 1 (Proof that Eq. (2.5) satisfies Mulholland's conditions)**.**
Regarding Point 1: Continuity and $f(0) = 0$ are clear. For monotonicity, note that $e^{qx} - 1$ is increasing, exponentiation is also increasing, and multiplying by $\alpha > 0$ preserves monotonicity. Thus, the composition is also increasing.
Regarding point 2: Similar reasoning applies for convexity. The composition of convex functions is convex, if the outer function is non-decreasing (Thomson et al., 2008, p. 481, Note 207). Since multiplication by a positive constant and exponentiation for exponents greater than 1 is both convex and increasing, the resulting function must be convex.

We now proceed to prove the convexity of $\log F(e^x)$. By logarithmic properties, it holds that

$$\log \alpha(e^{qe^x} - 1)^p = \log \alpha + p \cdot \log(e^{qe^x} - 1).$$

Arbitrary addition and multiplication by $p \geq 1$ preserve convexity, so it suffices to show the convexity of $\log(e^{qe^x} - 1)$. We prove this by examining the second derivative. If the second derivative is strictly positive, then the function is convex (Hug & Weil, 2020, p. 57, Remark 2.13). Differentiating twice yields

$$\frac{qe^{qe^x+x}\left(q(-e^x) + e^{qe^x} - 1\right)}{\left(e^{qe^x} - 1\right)^2} \overset{!}{>} 0.$$

The denominator is always positive and non-zero, so we can ignore it. Similarly, the factor qe^{qe^x+x} is always positive and can also be omitted. It remains to show that

$$-qe^x + e^{qe^x} - 1 > 0$$
$$\Updownarrow$$
$$e^{qe^x} - qe^x > 1$$

holds. This can be shown in two steps. First, we show that the left-hand side is strictly increasing. Second, we prove that its limit as $x \to -\infty$ is 1. Therefore, the inequality holds for all $x \in \mathbb{R}$.

1. For the first point, we show that the derivative is greater than zero. Differentiation yields

   $$qe^x \cdot e^{qe^x} - qe^x = qe^x \cdot (e^{qe^x} - 1) > 0.$$

   The inequality holds, since $e^x > 0$ for all $x \in \mathbb{R}$ and $e^x > 1$ for $x > 0$.

2. For the second point, we calculate the limit as $x \to -\infty$:

   $$\lim_{x\to-\infty} e^{qe^x} - qe^x = \lim_{x\to-\infty} e^{qe^x} - \lim_{x\to-\infty} qe^x = 1 - 0 = 1.$$

Thus, convexity is established, and the claim follows from Mulholland's Inequality.

□

Remark 2.3.3 (Minkowski metric is part of the p-exp metric family). If we set $\alpha = q^{-p}$ in (Eq. (2.5)), we obtain the Minkowski metric in the limit as $q \to 0$.

Proof. Setting $\alpha = q^{-p}$ yields

$$q^{-p}(e^{qx} - 1)^p = \left(\frac{e^{qx} - 1}{q}\right)^p.$$

As $q \to 0$, both the numerator and denominator tend to 0 (we can pull the limit inside the exponent since the exponentiation is continuous), therefore we can apply L'Hospital's rule. Hence,

$$\begin{aligned}
\lim_{q\to 0}\left(\frac{e^{qx} - 1}{q}\right)^p &= \left(\lim_{q\to 0}\frac{e^{qx} - 1}{q}\right)^p && \text{(continuity)}\\
&= \left(\lim_{q\to 0}\frac{x \cdot e^{qx} - 1}{1}\right)^p && \text{(L'Hôpital's rule)}\\
&= x^p.
\end{aligned}$$

□

Not every function that can be represented by the additive difference model is a metric.

Sources and Original Contributions

The theory presented in this chapter is mostly covered in the first four subsections(14.4.1-14.4.4) of Krantz et al. (1971, Chapter 14.4, pp. 175-185). In addition to the theory provided in this chapter, the authors also prove representation and uniqueness theorems for decomposability, interdimensional subtractivity and interdimensional additivity. Since some of these theorems require additional terminology, all three theorems have been omitted to focus on the AD Model.

All the axioms presented here relate to axioms of the structures, that have been the base of the reduction. Decomposability, intradimensional subtractivity and interdimensional additivity utilize reduction to a decomposable conjoint structure, absolute difference structure and an additive conjoint structure, respectively. These are presented in chapter 7, chapter 4.10 and chapter 6 in Krantz et al. (1989).

Other useful references include Beals et al. (1968), Tversky and Krantz (1970) and chapters 13, 15 and 16 in Breinbauer (1976).

Example 2.3.1 is stated as an exercise in Krantz et al. (1971, p. 79, Exercise 19) and its proof is provided by the author of this thesis. Remark 2.3.3 has been outlined in Krantz et al. (1971, p. 73).

Chapter 3

Segmental Additivity

In this chapter we will develop the theory for segmentally additive metrics. First we will define what segmentally additive metrics are and then develop segmentally additive proximity structures which lead to representation by such metrics. At the end we give some examples. This chapter is based on Busemann (1955, Chapter 1) and Chapters 12.3.1 and 14.2 from Krantz et al. (1971). A more detailed explanation of the sources can be found at the end of the chapter.

3.1 Segmentally Additive Metrics

As explained in Section 1.1, the goal is to find a metric representation in which, for any two points, there exists a curve connecting them such that the distances are additive – that is, the triangle inequality holds with equality. In order to establish this, we first introduce parametric curves, their length and rectifiable curves, which are curves with finite length. Since, at this point, rectifiable curves do not ensure additivity of distances, we introduce segments – rectifiable curves whose length is equal to the distance between the two points they connect. This guarantees the additivity of distances along the segment. Furthermore, we introduce the arc length, which is essentially the length of a sub-curve from the starting point to an intermediate point. Since this will be needed for a later proof, we define it here and provide a remark on it.

Definition 3.1.1 (Curves and Segments)**.**

1. Let δ be a metric on X. A *parametric curve* in $\langle X, \delta \rangle$ is a function γ that maps a closed interval $[a, b] \subset \mathbb{R}$ into X and is continuous. The parametric curve γ is said to join $\gamma(a)$ and $\gamma(b)$.

2. Let γ be a parametric curve in $\langle X, \delta \rangle$ with domain $[a, b]$. Its length $\mathrm{L}(\gamma)$ is the supremum of the sums

$$\sum_{i=1}^{n} \delta\left[\gamma(a_{i-1}), \gamma(a_i)\right]$$

over all finite sequences $a_0, a_1, \ldots, a_n$ such that

$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b.$$

If $\mathrm{L}(\gamma) < \infty$, then γ is *rectifiable.*

3. A rectifiable curve is a *segment* iff its length is equal to the distance between its endpoints, i.e., $\mathrm{L}(\gamma) = \delta\,(a, b)$.

4. For $a \leq x \leq b$ we denote the length of the sub-curve $\gamma(t),\ a \leq t \leq x$ from a to x by $\mathrm{L}_a^x(\gamma)$. This is also called the *arc length* of the curve.

The remark we provide on the arc length is intuitive and states that the arc length is continuous and non-decreasing. This will be useful in a later proof.

Remark 3.1.2 (Continuity & Monotonicity of Sub Length). For a rectifiable curve $\gamma(t),\ a \leq t \leq b$, the arc length $\mathrm{L}_a^x(\gamma)$ is a continuous, non-decreasing function of x.

Finally, we want all points in the metric space to be joined by a segment. Therefore, we define a metric space with additive segments where this property holds. An example for this is the Minkowski metric in $\mathbb{R}^n$ which we will demonstrate in Section 3.3.

Definition 3.1.3 (Metric Space with Additive Segments)**.**
Let δ be a metric on X. $\langle X, \delta \rangle$ is a *metric space with additive segments* iff for any $x, y \in X$ there is a segment joining them.

3.2 Segmentally Additive Proximity Structures

Now, we proceed with deriving segmentally additive proximity structures which ensure a metric representation with additive segments. We do this by examining necessary conditions for such a representation and translating them into ordinal criteria.

We start by observing the additivity of distances given three points x, y and z in a metric space with additive segments where y lies on the segment from x to y.

Then $\delta(x,y)+\delta(y,z)=\delta(x,z)$ holds. Suppose that x', y' and z' are any three points such that

$$\delta(x,y)\geq\delta(x',y') \tag{3.1}$$

and

$$\delta(x,z)\leq\delta(x',z'). \tag{3.2}$$

Then, by the triangle inequality and Equation (3.2), we have

$$\delta(x',y')+\delta(y',z')\geq\delta(x',z')\geq\delta(x,z)=\delta(x,y)+\delta(y,z).$$

Using Equation (3.1) on this equation yields

$$\delta(y',z')\geq\delta(y,z).$$

Further, if all the equalities are strict, the resulting inequality is also strict. As this property only includes inequalities, it can be restated in a purely ordinal criterion.

Definition 3.2.1 (Ternary Relation on Proximity Structures)**.**
Let $\langle\mathcal{A},\succsim\rangle$ be a factorial proximity structure. For $a, b, c\in\mathcal{A}$, the ternary relation $(a, b, c\rangle$ is said to hold iff for all $a', b', c'\in\mathcal{A}$:

1. If $(a, b)\succsim(a', b')$ and $(a', c')\succsim(a, c)$, then $(b', c')\succsim(b, c)$
2. If either of the preceding inequalities is strict, then the resulting inequality is also strict.

The relation $\langle abc\rangle$ is said to hold if both $(a, b, c\rangle$ and $(c, b, a\rangle$ hold.

This relation yields a notion of betweenness for the proximity structure in terms of the relation $\succsim$. We will briefly discuss this relation and its implications. It is practical to examine this relation in terms of isosimilarity contours. Intuitively, we have a circle around x with radius $\delta(x,z)$. The set of all points at a distance $\delta(x,z)$ to x forms the isosimilarity contour for this distance. Using this notion, we can observe that y minimizes the distance to z on the given sphere where y resides. To see that, let y' be such that $\delta(x,y)=\delta(x,y')$. Using the triangle inequality and the additivity yields

$$\delta(x,y)+\delta(y,z)=\delta(x,z)\leq\delta(x,y')+\delta(y',z).$$

By assumption, $\delta(x,y)=\delta(x,y')$ holds, and therefore $\delta(y,z)\leq\delta(y',z)$, which means that the minimal distance is the difference in the radii.

Moreover, we can observe similarly, that given another point $\tilde{z}$ with $\delta(x,z) = \delta(x,\tilde{z})$, the point $\tilde{y}$ with minimal distance to $\tilde{z}$ has the same distance to $\tilde{z} - \delta(\tilde{y},\tilde{z}) = \delta(y,z)$. Stated differently, this assures that the minimal distance is independent of the direction, i.e., independent of the choice of z. Put together with the fact that x is arbitrary, this yields that the minimized distance is independent of the direction (choice of z) and the location of the sphere (choice of x) and only depends on the two radii. These necessary conditions we just derived are visualized in Figure 3.1.

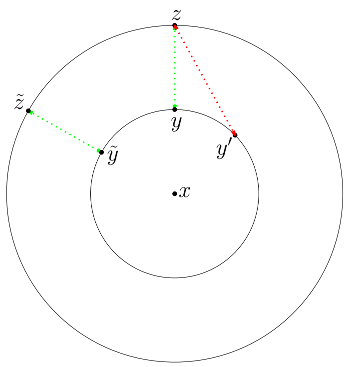

Figure 3.1: Illustration of betweenness with isosimilarity contours.
For an explanation, see the text above. Green lines represent minimal distances, white red lines indicate that the distance is not minimal. The isosimilarity curves are illustrated intuitively as circles; however, this is generally not the case.

Since we represent the distance with a metric with additive segments, for every $\alpha \leq \delta(x,z)$, there is a point y on the segment from x to z such that $\delta(x,y) = \alpha$. Thus, a necessary condition for a metric representation with additive segments is this condition reformulated for proximity structures.

Axiom 3.2.2 (Segmental Solvability)**.**
A proximity structure $\langle \mathcal{A}, \succsim \rangle$, with a ternary relation $\langle\rangle$ defined in terms of $\succsim$, satisfies *segmental solvability* iff for all $a, c, e, f \in \mathcal{A}$ with $(a,c) \succsim (e,f)$, there exists $b \in \mathcal{A}$ such that $(a,b) \sim (e,f)$ and $\langle abc \rangle$.

In this definition, the distance α is represented by (e,f) and the minimality is assured by the ternary betweenness relation.

The next step is to construct the distances along the segments. The basic idea is the following: We wish to approximate the ratio of $\delta(a,b)$ and $\delta(c,d)$. This is done by choosing a common distance for the steps and using segmental solvability to iteratively partition the segments into subsegments of the common distance. This process yields two sequences, $a = a_0, a_1, \ldots$ and $c = c_0, c_1 \ldots$ of which all terms are between the start and end point, a and b or c and d, respectively. If this construction terminates, and we denote p the steps needed for a_i to reach $\mathscr{b}$ and q the steps for c_i to reach $\mathscr{d}$, then the ratio $\frac{\delta(a,\mathscr{b})}{\delta(c,\mathscr{d})}$ is approximately $\frac{p}{q}$.

Since, at this point, we cannot ensure that this process terminates, we introduce another axiom, the Archimedean or connectedness axiom, to guarantee that it does. This axiom states that there is no "infinitely small" distance that would allow the process to be executed infinitely.

Putting all together we get the following definition:

Definition 3.2.3 (Segmentally Additive Proximity Structure)**.**
A proximity structure $\langle \mathcal{A}, \succsim \rangle$, with a ternary relation $\langle \rangle$ defined in terms of $\succsim$, is segmentally additive if and only if the following two axioms hold for all $a, c, e, f \in \mathcal{A}$:

1. Segmental solvability
2. If $e \neq f$, then there exist $\mathscr{b}_0, \ldots, \mathscr{b}_n \in \mathcal{A}$, such that $\mathscr{b}_0 = a$, $\mathscr{b}_n = c$, and $(e, f) \succsim (\mathscr{b}_{i-1}, \mathscr{b}_i)$ for $i = 1, \ldots, n$. (Archimedean axiom)

A segmentally additive proximity structure is complete if and only if the following two axioms hold for all $a, c, e, f \in \mathcal{A}$:

3. If $e \neq f$, then there exist $\mathscr{b}, \mathscr{d} \in \mathcal{A}$, with $\mathscr{b} \neq \mathscr{d}$, such that $(e, f) \succ (\mathscr{b}, \mathscr{d})$. (nondiscreteness of distances)
4. Let $a_1, \ldots, a_n, \ldots$ be a sequence of elements of $\mathcal{A}$, such that for all $e, f \in \mathcal{A}$, if $e \neq f$, then $(e, f) \succsim (a_i, a_j)$ for all but finitely many pairs (i, j). Then there exists $\mathscr{b} \in \mathcal{A}$, such that for all $e, f \in \mathcal{A}$, if $e \neq f$, then $(e, f) \succsim (a_i, \mathscr{b})$ for all but finitely many i. (Limits for Cauchy sequences)

A segmentally additive proximity structure provides a metric representation with additive segments, see Krantz et al. (1971, p. 168). However, as our goal is to establish a representation by the Minkowski metric, we need a representation in a complete metric space with additive segments. For this, two further conditions must be added. The first one is the nondiscreteness of distances which assures that there is no smallest distance. The second one is the familiar notion of convergence of Cauchy sequences, which assures that the underlying space is complete. Here (e, f) corresponds to ε and $\mathscr{b}$ to the limit point of the sequence a_i. These 4 conditions are both sufficient and necessary for such a representation, see Krantz et al. (1971, p. 168).

Theorem 3.2.4 (Representation and Uniqueness Theorem for Complete Segmentally Additive Proximity Structures)**.**

Suppose $\langle \mathcal{A}, \succsim \rangle$ is a complete segmentally additive proximity structure. Then $\langle \mathcal{A}, \succsim \rangle$ is representable by a metric, and additionally, for all $a, b, c \in \mathcal{A}$:

1. *$\langle abc \rangle$ iff $\delta(a,b) + \delta(b,c) = \delta(a,c)$.*

2. *δ' is a ratio scale.*

Furthermore, $\langle \mathcal{A}, \delta \rangle$ is a complete metric space with additive segments.

3.3 Examples

Example 3.3.1 (Segmental Additivity of the Minkowski Metric)**.**
For $p \geq 1$ the Minkowski Metric is a metric with additive segments[1]. The segments are all straight lines (i.e., the line segments between two arbitrary points).

Proof. Let $\mathbf{x}, \mathbf{z} \in \mathbb{R}^n$ and $0 \leq t \leq 1$. Then, for $\mathbf{y} = (1-t)\mathbf{x} + t\mathbf{z}$ ($\mathbf{y}$ is on the line segment between $\mathbf{z}$ and $\mathbf{x}$, iff $\mathbf{y}$ can be represented in this way) it holds that:

$$\begin{aligned}
\delta_p(\mathbf{x}, \mathbf{y}) + \delta_p(\mathbf{y}, \mathbf{z}) &= \left[\sum_{i=1}^{n} |x_i - (1-t)x_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |(1-t)x_i + tz_i - z_i|^p\right]^{1/p} \\
&= \left[\sum_{i=1}^{n} |x_i - x_i + tx_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |x_i - tx_i + tz_i - z_i|^p\right]^{1/p} \\
&= t\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} + (1-t)\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} \\
&= \delta_p(\mathbf{x}, \mathbf{z}).
\end{aligned}$$

□

Sources and Original Contributions

The section on segmentally additive metrics is shortly summarized in Krantz et al. (1971, Chapter 12.3.1, pp. 52-57). A more detailed introduction to metric geometry is Busemann (1955, Chapter 1, pp. 1-64), where curves and segments are the basis for the theory developed afterward. The remark on arc-length is only stated in Busemann (1955, p. 21).
For the section on segmentally additive proximity structures, the main reference was Krantz et al. (1971, Chapter 14.2, pp. 163-174). In addition to the content provided here, the methods to estimate distances are further explained, and the provided

[1]This is shorthand for: $\mathbb{R}^n$ with the Minkowski metric is a metric space with additive segments.

theorems are proved. Additional references discussing segmentally additive metrics in the context of proximity structure include Tversky and Krantz (1970) and Beals et al. (1968). Example 3.3.1 can be found in Tversky and Krantz (1970, p. 588).

Chapter 4

Combining Segmental Additivity with the AD Model

In this chapter we aim to connect the previous two chapters and working with proximity structures that are both, segmentally additive and AD structures. We will present and prove the two main results. The first one are sufficient and necessary conditions for the AD model to be a metric that is additive on the coordinate axes. The second one is the unique representation theorem which states that the only additive difference model that is compatible with a metric with additive segments is the Minkowski metric. We will prove both theorems, specifically extending and completing the proof of the second theorem, as the proofs presented in Krantz et al. (1971, pp. 192-199) are incomplete in some aspects. As in the previous chapters, the sources are further explained at the end of this chapter.

4.1 Conditions for Metric Representation of AD Structures

We begin by combining additive difference structures with a metric. However, since metrics are too general, we restrict our focus to a specific class of metric – those that are additive on the coordinate axes. This leads to necessary and sufficient conditions, which are stated in the following theorem.

Theorem 4.1.1 (Conditions for Metric Representation of AD Structures).
Suppose $\langle \mathcal{A}, \succsim \rangle$ is an additive-difference proximity structure such that f is the

additive-difference representation from Theorem 2.2.7, i.e.,

$$f(a,b) = \sum_{i=1}^{n} F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \ldots, n$ (through transformation, since F_i are interval scales).

For $\langle \mathcal{A}, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes, it is

(i) necessary that there exists a continuous, convex function F, such that $F_i(x) = F(t_i x), t_i > 0$, for all x in the domain of F_i, $i = 1, \ldots, n$; and

(ii) sufficient that, in addition (to the conditions in (i)), the function $\log F(e^x)$ is convex.

Before beginning with the proof we remark what additivity on the coordinate axes means for elements in the proximity structure.

Remark 4.1.2 (Additivity on the Coordinate Axes). Additivity on the coordinate axes means the following: If $a, b, c \in \mathcal{A}$ differ only on one factor $\mathcal{A}_i$, then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a,c) \succsim (a,b), (b,c)$$

implies that $\delta(a,b) + \delta(b,c) = \delta(a,c)$. This can be also formulated in terms of betweenness (Axiom 2.2.2). If a, b and c differ only on one factor and $a|b|c$, then $\delta(a,b) + \delta(b,c) = \delta(a,c)$.

In this section, until the proof of this theorem is finished, everything will be assumed as in Theorem 4.1.1, unless otherwise stated. Now we proceed with the proof of the theorem.

Proof of Theorem 4.1.1 (i). According to the assumptions of the theorem, the additive difference representation $f(a,b)$ is compatible with a metric δ on $\mathcal{A}$, i.e.,

$$f(a,b) \leq f(c,d) \quad \text{iff} \quad \delta(a,b) \leq \delta(c,d) \quad \text{for all } a,b,c,d \in \mathcal{A}.$$

Therefore, there exists an increasing function G[1], such that for all $a, b \in \mathcal{A}$

$$\delta(a,b) = G(f(a,b)) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right).$$

[1] A possible perspective on this is, that the metric δ induces an ordinal scale for pairs of elements. The same holds for the function f and these orderings are the same. Because ordinal scales, by definition, are invariant under increasing bijections, there must exist an increasing function G, which transforms the f-scale into the δ-scale.

Assume a, b, and c differ only in the i-th factor and $a|b|c$. Thus, we can use additivity on the i-th coordinate axis for $a, b, c \in \mathcal{A}$ which yields

$$G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) + G\left(F_i\left(|\varphi_i(b_i) - \varphi_i(c_i)|\right)\right) = G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(c_i)|\right)\right).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x+y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x+y)$ for all x, y in the domain of $F_i, 1 \leq i \leq n$. By Theorem 2.2.7, the domain is $\mathbb{R}$. This is Cauchy's functional equation and, under the assumption of monotonicity, the only solution is $H_i = t_i x$ for $t_i \in \mathbb{R}$ (Aczel & Dhombres, 1989, p. 15). Since the values are positive (as we are dealing with metrics), $H_i = t_i x$ with $t_i > 0$ must hold for all $i = 1, ..., n$. Additionally, since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i, the F_i can only differ in range and in the unit of the domain (i.e., the scaling factor t_i which defines the domain of F_i by adjusting the domain of G^{-1}). Without loss of generality, we can assume that the range of F_n is the largest among the F_i. This assumption allows us to define $F = F_n$ as the reference function, and the other F_i can be expressed as $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i.
Continuity and convexity will be proven in the next lemmas. After that, we proceed with sufficiency. □

Lemma 4.1.3.

$$F(x+y+z) + F(z) \geq F(x+z) + F(y+z).$$

Proof. In the triangle inequality,

$$F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(y_i)\right] \geq F^{-1}\left[\sum_{i=1}^{n} F(x_i + y_i)\right],$$

let $x_1 = z, x_2 = F^{-1}(F(x+z) - F(z)), x_j = 0$ for $3 \leq j \leq n$ and $y_1 = y, y_j = 0$ for

$2 \leq j \leq n$. Hence,

$$\begin{aligned} F(x+y+z) &= F\left[F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(y_i)\right]\right] \\ &\geq \sum_{i=1}^{n} F(x_i + y_i) \qquad \text{(by the triangle inequality)} \\ &= F(y+z) + F(x+z) - F(z). \end{aligned}$$

□

Lemma 4.1.4.
F is continuous.

Proof. Since F is increasing and defined on an interval, all discontinuities are gaps (Thomson et al., 2008, p. 335)[2]. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this. Assume $F(x)$ is the lower boundary of a gap, then there exists $\varepsilon > 0$ such that for all $\delta > 0$[3], it must hold that $F(x+\delta) - F(x) > \varepsilon$. However, by Lemma 4.1.3, for every $y > 0$,

$$F(x+y+\delta) - F(x+y) \geq F(x+\delta) - F(x) > \varepsilon,$$

and so there is a gap at any $x+y$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. □

Lemma 4.1.5 (Characterization of Convexity)**.**
Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $x, y, z \geq 0$ the inequality from Lemma 4.1.3 holds, i.e.,

$$F(x+y+z) + F(z) \geq F(x+z) + F(y+z).$$

Proof.

[2](Thomson et al., 2008) prove the following statement: Let f be a real-valued monotonic function defined on an interval $[a, b]$. Then the set of points of discontinuity of f in that interval is countable. Combining this statement with the paragraph above it, yields that these discontinuities must be gaps. Since any interval (open, half-open and unbounded intervals) can be represented by a countable union of closed intervals, this theorem can be extended to arbitrary intervals since countable unions of countable sets are still countable. This can be done by using $(a, b) = \bigcup_{n=1}^{\infty}[a + 1/n, b - 1/n]$, or $(a, \infty) = \bigcup_{n=1}^{\infty}[a + 1/n, a + n]$. Similar arguments apply to other types of intervals.

[3]Here, δ differs from our usual usage and is not a metric; rather it follows the standard notation for continuity and is only used in this proof this way.

Part 1 ($\Rightarrow$)**.**
Suppose, F is convex, and let $\lambda = x/(x+y)$, then

$$\begin{aligned}\lambda(x+y+z) + (1-\lambda)z &= x+z\\ (1-\lambda)(x+y+z) + \lambda z &= y+z.\end{aligned}$$

Hence, by convexity,

$$\begin{aligned}F(x+z) + F(y+z) &= F\left[\lambda(x+y+z) + (1-\lambda)z\right] + F\left[(1-\lambda)(x+y+z) + \lambda z\right]\\ &\leq \lambda F(x+y+z) + (1-\lambda)F(z) + (1-\lambda)F(x+y+z) + \lambda F(z)\\ &= F(x+y+z) + F(z).\end{aligned}$$

Part 2 ($\Leftarrow$)**.**
To prove the converse, we will use the continuity of F by Lemma 4.1.4. Suppose, $x > y$, then

$$\begin{aligned}F(x) + F(y) &= F\left(\frac{x-y}{2} + \frac{x-y}{2} + y\right) + F(y)\\ &\geq 2F\left(\frac{x-y}{2} + y\right) && \text{(by Lemma 4.1.4)}\\ &= 2F\left(\frac{x+y}{2}\right),\end{aligned}$$

which implies convexity for a continuous F (Hug & Weil, 2020, p. 51, Exercise 6).

□

We proceed with the proof of sufficiency.

Proof of Theorem 4.1.1 (ii). Sufficiency is established by the Mulholland Theorem, which was already used in Section 2.3. By this theorem, the triangle inequality holds for the AD model, and therefore, the model defines a metric. The proofs of symmetry and positivity are identical to those presented in Section 2.3. □

We conclude this section by discussing the implications of this theorem for the form of the metric representation. This observation will be important in the next chapter and ties this result closer to the Minkowski metric.

Remark 4.1.6 (Form of δ). Examining the proof of Theorem 4.1.1 more closely yields, that $G = F^{-1}$ must hold and the representation of δ takes the form

$$\delta(a, b) = F^{-1}\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\}$$

where F is continuous, convex, and increasing. This follows from two arguments.

Firstly, incorporating the t_i into the φ_i (which is possible because, by Theorem 2.2.7, the φ_i are interval scales)

$$\begin{aligned} G\Big\{\sum_{i=1}^{n} F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} &= G\Big\{\sum_{i=1}^{n} F(t_i|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \\ &= G\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \end{aligned}$$

yields the desired form inside the sum.

Secondly, by using $F_i(x) = F_n(t_i x)$ and $F_i(x) = G^{-1}(t_i x)$. The first equation applied for $i = n$ yields $F_n(x) = F_n(t_n x)$ which implies $t_n = 1$. Using this in the second equation for $i = n$ yields $F(x) = F_n(x) = G^{-1}(t_n x) = G^{-1}(x)$. Hence, by taking the inverse on both sides (which is possible due to F and G both being increasing), we get the desired form for δ.

4.2 Unique Representation Theorem

In this section, we present the main theorem and complete the proofs given by Krantz et al. (1971, pp. 195-199). The remainder of this chapter, except for the final section, consists of the proof of this theorem. The last section is dedicated to proving an intermediate result from functional analysis that is used in the proof. We begin by stating the theorem, and follow a top-down approach in the proof to provide a broad understanding. This means, we use results established in Section 4.2.1. The same approach applies to the proofs in that section; see Section 4.2.1 for further explanation. We start with the statement of the theorem.

Theorem 4.2.1 (Unique Representation Theorem: The Power Metric as the Sole Representation for Segmentally Additive AD Structures)**.**
Suppose $\langle \mathcal{A}, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure. Then there exists a unique $p \geq 1$ and real-valued functions φ_i, defined on $\mathcal{A}_i$, $i = 1, \ldots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in \mathcal{A}$,

$$(a, b) \succsim (c, d) \quad \textit{iff} \quad \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^p\right]^{1/p}.$$

Proof. This proof proceeds in multiple steps. First, we make observations about the domains and ranges of the functions F_i, use this to find bounds for the domain of F. Using these bounds, we argue that F can be extended to the positive reals. Next, we derive from this a functional equation, that is only satisfied by $f(x) = \alpha x^p$. Finally, we prove that F is equal to this extension on the whole domain, and therefore δ is a Minkowski metric.

Part 0 (Establishing the basics and using the Theorem 4.2.3)**.**
By Theorem 4.1.1 and Remark 4.1.6, δ must be of the form

$$\delta\left(a, b\right) = F^{-1}\left[\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\right]$$

where F is continuous, convex, and increasing. By the same argument as in Remark 4.2.4 (iii), δ must be defined on a Cartesian product of intervals of the form $[0, \rho_i)$ where ρ_i are the ranges of $|\varphi_i(a_i) - \varphi_i(b_i)|$ for $i = 1, \dots, n$. This Cartesian product is convex and, by leaving out 0 in each product, also open. Therefore, the requirements for Theorem 4.2.3 are met if we leave out $\mathbf{0}$ and applying the lemma yields homogeneity for $\mathbf{x}, \mathbf{z} \neq \mathbf{0}$:

$$\delta\left(\mathbf{x}, \mathbf{x} + t(\mathbf{z} - \mathbf{x})\right) = t\,\delta\left(\mathbf{x}, \mathbf{z}\right).$$

Applying homogeneity for the definition of δ above and setting $x_i = |\varphi_i(b_i) - \varphi_i(a_i)|$ yields

$$F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] = t \cdot F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right]. \tag{4.1}$$

However, due to translation invariance, homogeneity also holds for $\mathbf{x} = \mathbf{0}$[4]. We will show, that the only solutions to this equation are of the form $F(x) = \alpha x^p$, i.e., the power functions.

First, we start by observing the domains and the ranges of F. By Theorem 4.1.1, the additive difference model

$$\delta(a, b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(x) = F(t_i x)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)$,

[4]By adding $0 = \mathbf{y} - \mathbf{y}$ for some $\mathbf{y}$ in the domain, we get homogeneity for $\mathbf{0}$:

$$\delta\left(\mathbf{0}, t\mathbf{z}\right) = F^{-1}\left[\sum_{i=1}^{n} F(t(y_i - (y_i - z_i)))\right] = tF^{-1}\left[\sum_{i=1}^{n} F(y_i - (y_i - z_i))\right] = t\,\delta\left(\mathbf{0}, \mathbf{z}\right).$$

and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain). Since the domains of F_i are intervals (because δ is a metric with additive segments and therefore for any point in the domain of F_i, the interval between zero and that point must also be in the domain), we can choose $\omega > 0$ such that for each $i = 1, \ldots, n$ and any x with $0 \leq x \leq \omega/t_i$, x lies within the domain of F_i. In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1}\left[\sum_{i=1}^{n} F_i(\omega_i)\right] = F^{-1}\left[nF(\omega)\right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 1 (Extending the bounds of F)**.**
For $x < \omega$, we have by Equation (4.1),

$$\begin{aligned} F^{-1}[nF(x)] &= F^{-1}[nF(x\omega/\omega)] \\ &= (x/\omega)F^{-1}[nF(\omega)] \\ &= x\Omega/\omega. \end{aligned}$$

Thus, F satisfies the following, by plugging in the correct values for x ($x\frac{\omega}{\Omega}$ and $F^{-1}(\frac{x}{n})$)

$$F(x) = nF(x\omega/\Omega), \quad 0 \leq x \leq \Omega; \tag{4.2}$$

$$F^{-1}(x) = (\Omega/\omega)F^{-1}(x/n), \quad 0 \leq x \leq nF(\omega). \tag{4.3}$$

These two equations can now be used to extend the domain of F and F^{-1}.

We use Equation (4.2) by defining F through the right-hand side (for values where the right-hand side is defined, i.e., for $0 \leq x \leq \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this indeed extends F beyond the interval $[0, \Omega]$. Similarly, F^{-1} satisfies Equation (4.3) for the extended interval $0 \leq x \leq nF(\Omega) = n^2F(\omega)$. Moreover, Equation (4.1) is now valid for $0 \leq t \leq 1$ and $0 \leq x_i \leq \Omega$:

$$\begin{aligned} F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(tx_i\omega/\Omega)\right] && \text{(by Eq. (4.2))} \\ &= F^{-1}\left[n\sum_{i=1}^{n} F(tx_i\omega/\Omega)\right] \\ &= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(tx_i\omega/\Omega)\right] && \text{(by Eq. (4.3))} \\ &= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(x_i\omega/\Omega)\right] \\ & \text{(by homogeneity applied for } x_i\omega/\Omega \leq \omega) \end{aligned}$$

$$
\begin{aligned}
&= tF^{-1}\left[n\sum_{i=1}^{n} F(x_i\omega/\Omega)\right] && \text{(by Eq. (4.3))}\\
&= tF^{-1}\left[\sum_{i=1}^{n} nF(x_i\omega/\Omega)\right]\\
&= tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right]. && \text{(by Eq. (4.2))}
\end{aligned}
$$

Further, the extended function is both increasing and continuous since the extension is based on scaling of the domain and the range of the original function.

For continuity let $x_k \to x$ with $\omega < x < \Omega$. Then

$$
\begin{aligned}
\lim_{k\to\infty} F(x_k) &= \lim_{k\to\infty} nF\left(x_k\frac{\omega}{\Omega}\right) && \text{(Definition of } F(t) \text{ for } \omega \le t \le \Omega)\\
&= nF\left(\lim_{k\to\infty} x_k\frac{\omega}{\Omega}\right) && \text{(Continuity of } F(t) \text{ for } 0 \le t \le \omega)\\
&= nF\left(x\frac{\omega}{\Omega}\right) && \text{(Convergence of } x_k)\\
&= F(x) && \text{(Definition of } F(t) \text{ for } \omega \le t \le \Omega)
\end{aligned}
$$

which is equivalent to F begin continuous on the extended domain.

For monotonicity, we can use a similar argument. Let $x < x'$ with $\omega < x, x' < \Omega$. Then $x\frac{\omega}{\Omega} < x'\frac{\omega}{\Omega}$ holds. Plugging that into F and multiplying by n yields $nF\left(x\frac{\omega}{\Omega}\right) < nF\left(x'\frac{\omega}{\Omega}\right)$ due to F being increasing on the original domain. Therefore, $x < x'$ implies $F(x) < F(x')$ which is equivalent to F increasing. Thus, we have established strict increasing monotonicity, continuity and Equation (4.1) for the extended function.

Repeatedly applying the arguments, extends the intervals in which F is defined and in which monotonicity, continuity, and homogeneity (Eq. (4.1)) are valid by a factor of Ω/ω each time. Thus, we can extend F to $[0, \infty)$ while preserving these properties.

We can prove further, that the homogeneity equation must also hold for $t \in [0, \infty)$. For that, let $x_i > 0$ and $t > 1$. Then there exists $t' = \frac{1}{t} < 1$ such that $t' \cdot tx_i = x_i$. Therefore,

$$
\begin{aligned}
tt'F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] && (tt' = 1)\\
&= F^{-1}\left[\sum_{i=1}^{n} F(tt'x_i)\right] && (tt' = 1)\\
&= t'F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right]
\end{aligned}
$$

The last step holds, because both $t' < 1$ and $tx_i < \infty$ hold. Dividing both sides by t' yields

$$F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] \qquad \text{for } x, t < \infty.$$

To sum up, we have obtained an extended function which satisfies monotonicity, continuity, and homogeneity for $t \in [0, \infty)$. This extended function must coincide with tie original F on at least the interval $[0, \Omega]$ (because this was the domain of F that we deduced, before extending it).

Part 2 (Deriving functional equation)**.**
Next, we derive a functional equation that allows us to use results on quasi-arithmetic homogeneous means. The theorem states as follows:

Theorem (Uniqueness of homogeneous quasi-arithmetic means, Theorem 4.2.19)**.** *Suppose that ϕ is continuous and strictly increasing in the open interval $(0, \infty)$ and that*

$$\mathcal{M}_\phi(t\mathbf{x},\ \mathbf{w}) = t\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) \quad \textit{with} \quad \mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i \phi(x_i)\right)$$

for all positive $\mathbf{x}, \mathbf{w}, t$ and all $n \in \mathbb{N}$ such that $\sum_{i=1}^{n} w_i = 1$. Then $\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \mathcal{M}_p(\mathbf{x}, \mathbf{w})$. (i.e., $\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$ is either a power mean with $\phi(x) = \alpha x^p + \beta$ and $\alpha, \beta, p \in \mathbb{R}, p > 0$ or a geometric mean with $\phi(x) = \alpha \log(x)$, $\alpha \in \mathbb{R}$.)

The proof of this statement will be presented in Section 4.2.2, here we will apply the theorem for our proof.

Since all requirements for ϕ are already met by F, we have to derive the functional equation for all $n \in \mathbb{N}$ and all positive weights. First, we will prove that the homogeneity holds not only for sums of n summands, but for arbitrary many summands $N \in \mathbb{N}$. Then we will incorporate equal weights into this equation, extend them to commensurable weights (i.e., rational weights with a common denominator) and then use this to approximate arbitrary real weights.

Subpart 2.1 (Homogeneity for arbitrary long sums)**.**
Let for $N < n$ the first N coordinates of $\mathbf{x}$ be zero, i.e., $\mathbf{x} = (x_1, \dots, x_N, 0, \dots, 0)$. Then, since $F(0) = 0$ it holds that, $F(x_i) = 0$ for $i > N$. Applying homogeneity for

this $\mathbf{x}$ yields

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] && (x_i = 0 \text{ for } i > N) \\
&= tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] && \text{(homogeneity)} \\
&= tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right]. && (x_i = 0 \text{ for } i > N)
\end{aligned}$$

Therefore, we have homogeneity for $N \leq n$:

$$F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right]. \tag{4.4}$$

Now, we proceed with homogeneity for $n+1$ summands. This can be achieved by grouping the last two summands into one and applying homogeneity twice:

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{n+1} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(tx_n) + F(tx_{n+1})\right] && \text{(expanding sum)} \\
&= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(F^{-1}(F(tx_n) + F(tx_{n+1})))\right] \\
& && \text{(regrouping)} \\
&= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(tF^{-1}(F(x_n) + F(x_{n+1})))\right] \\
& && \text{(Eq. (4.4) on the last summand, } N = 2) \\
&= tF^{-1}\left[\sum_{i=1}^{n-1} F(x_i) + F(F^{-1}(F(x_n) + F(x_{n+1})))\right] \\
& && \text{(homogeneity for } n \text{ summands)} \\
&= tF^{-1}\left[\sum_{i=1}^{n-1} F(x_i) + F(x_n) + F(x_{n+1})\right] && \text{(cancellation of inverses)} \\
&= tF^{-1}\left[\sum_{i=1}^{n+1} F(x_i)\right].
\end{aligned}$$

Therefore, we now have homogeneity for $n+1$ summands. Repeating this, yields homogeneity for arbitrary many summands:

$$F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right] \quad N \in \mathbb{N}. \tag{4.5}$$

Subpart 2.2 (Homogeneity with weights)**.**
Now, we proceed with incorporating weights into the equation. Firstly, applying Equation (4.3) on the left-hand side of Equation (4.5) yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N}F(tx_i)\right]=F^{-1}\left[\sum_{i=1}^{N}F(tx_i)\right]=tF^{-1}\left[\sum_{i=1}^{N}F(x_i)\right].$$

Applying Equation (4.3) once more on the right-hand side, yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N}F(tx_i)\right]=F^{-1}\left[\sum_{i=1}^{N}F(tx_i)\right]=tF^{-1}\left[\sum_{i=1}^{N}F(x_i)\right]=\Omega/\omega tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{N}F(x_i)\right].$$

Hence,

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N}F(tx_i)\right]=tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{N}F(x_i)\right]$$

must hold for all $N\in\mathbb{N}$. Repeatedly applying this argument yields that

$$F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N}F(tx_i)\right]=tF^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N}F(x_i)\right] \tag{4.6}$$

must hold for all $l,N\in\mathbb{N}$.

Now we extend this to weights of the form $\frac{q}{n^l}$ with $q,l\in\mathbb{N}$. For this, denote the set of all these numbers as $\mathbb{W}:=\{\frac{q}{n^l}:q,l\in\mathbb{N}\}$. Let $\mathbf{w}\in\mathbb{W}^N$. Then we can write $w_i=\frac{q_i}{n^{l_i}}$ for all $i=1,\dots,N$. We can assume that l_i is the same for all i without loss of generality, by having n_l be a common denominator and expanding all fractions accordingly. Therefore, we not have weights $w_i=\frac{q_i}{n^l}$ for $i=1,\dots N$. The basic idea now is, to expand each summand $q_i\cdot F(tx_i)$ into a sum of $F(tx_i)$ with length q_i. For that, let $Q=\sum_{i=1}^{N}q_i$ and define $\tilde{x}_i$ by repeating x_i for q_i times:

$$\tilde{x}_i=\begin{cases}x_1 & i=1,\dots,q_1\\ x_2 & i=q_1+1,\dots,q_2\\ & \vdots\\ x_N & i=q_{N-1}+1,\dots,q_N\end{cases}.$$

Then

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{N} \frac{q_i}{n^l} F(tx_i)\right] && \text{(plugging in } w_i = \tfrac{q_i}{n^l}\text{)}\\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N} q_i F(tx_i)\right] && \text{(rearranging)}\\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N}\sum_{j=1}^{q_i} F(tx_i)\right] && (q_i \in \mathbb{N})\\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{Q} F(t\tilde{x}_i)\right] && \text{(Definition of } \tilde{x}_i\text{)}\\
&= tF^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{Q} F(\tilde{x}_i)\right] && \text{(Eq. (4.6))}\\
&= tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right] && \text{(reversing the steps)}
\end{aligned}$$

Thus, we have derived

$$F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right] \quad w_i \in \mathbb{W}. \tag{4.7}$$

Now, in order to approximate real weights, we need density of $\mathbb{W}$ in $\mathbb{R}_{\geq 0}$. Since n^l gets arbitrarily small, there exists for every $\varepsilon > 0$ an l such that $n^l < \varepsilon$. Therefore, if $x - y < \varepsilon$, for $x, y \in \mathbb{R}_{\geq 0}$ and $x > y$ there exists $k \in \mathbb{N}$ such that $x < \frac{k}{n^l} < y$. Thus, $\mathbb{W}$ is dense in $\mathbb{R}_{\geq 0}$, and we can approximate real weights with converging sequences from $\mathbb{W}$.

Finally, let $w_i \in \mathbb{R}_{\geq 0}$ and $q_{i_k} \in \mathbb{W}$ be sequences for $i = 1, \ldots, N$ such that $q_{i_k} \to w_i$ for $k \to \infty$. Then

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{N} \lim_{k\to\infty} q_{i_k} F(tx_i)\right] && \text{(Convergence of } q_{i_k}\text{)}\\
&= \lim_{k\to\infty} F^{-1}\left[\sum_{i=1}^{N} q_{i_k} F(tx_i)\right] && (F^{-1} \text{ continuous } ^5)\\
&= \lim_{k\to\infty} tF^{-1}\left[\sum_{i=1}^{N} q_{i_k} F(x_i)\right] && \text{(Eq. (4.7))}\\
&= tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right]. && \text{(reversing the steps)}
\end{aligned}$$

Hence, the homogeneity holds for all $x_i, w_i, t \in \mathbb{R}_{\geq 0}$ and $N \in \mathbb{N}$, in particular, it holds for weights with sum 1. By Theorem 4.2.19 the only monotonic solutions are the power functions $\alpha x^p + \beta$ or the logarithm $\alpha \log(x)$ with $\alpha, \beta, p \in \mathbb{R}$ and $p > 0$. However, since $F(x) \geq 0$, it cannot be the logarithm and by $F(0) = 0$, the vertical shift β must be equal to 0. Therefore, the only solution is, that the extended function takes the form αx^p, since we applied this theorem on the extension of F. Therefore, $F(x) = \alpha x^p$ must hold at least on the interval $[0, \Omega)$.

Part 3 (Proving equality for positive reals)**.**
Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega$ for $i = 1, \dots, n$. Applying monotonicity yields (we can assume $t \leq 1$):

$$\begin{aligned}
\delta(a, b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t} F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^p\right)^{1/p} && (\text{ since } t|\varphi_i(a_i) - \varphi_i(b_i)| \leq \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^p\right)^{1/p} \\
&= \delta_p\left(\varphi_i(a_i), \varphi_i(b_i)\right)
\end{aligned}$$

Thus, δ is a Minkowski metric and, by Theorem 2.2.7, the F_i are interval scales, allowing for affine transformations. However, since we set $F_i(0) = 0$, the affinity vanishes, and we are left with $F_i(x) = \alpha \cdot x^p$. Finally, the restriction $1 \leq p < \infty$ follows from Minkowski's inequality (Bullen, 2003, pp. 189-191 Theorem 9).

□

4.2.1 Homogeneity Theorem

Before beginning this section, we will first introduce some notation.
δ_∞ is used to represent the ∞-metric (also called the maximum metric) which is induced by the maximum norm[6]. This metric induces the natural topology of the Euclidean space. $\mathrm{B}^{\delta}_{(r)} := \{x \in \mathbb{R}^n : \delta(x) < r\}$ denotes the open Ball with radius r around 0 with respect to the metric δ; similarly $\mathrm{B}^{\delta}_{[r]}$ denotes the closed ball. We use ∞ as a subscript, to indicate that the ball is with respect to δ_∞.

In this section we will prove the homogeneity theorem, which is needed for the proof of the main result, Theorem 4.2.1. The proof is divided into 5 parts and

[6]The maximum norm is defined by $\|x, y\|_\infty := \max_{i \leq n}\{|x_i - y_i|\}$.

a whole subsection which is devoted to showing the convexity of δ-balls which is needed for the proof. First, we will state the theorem, give some remarks on it and prove some lemmas. Then we will prove the homogeneity theorem. Everything in this section is assumed as in this theorem. For example, if we write δ, this implies not any metric but the metric from this theorem with its properties. Moreover, all topological terms (open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e., the topology induced by the δ_∞ and the Euclidean metric), not the topology induced by δ or the induced topology on a subset S. This will be especially important in the subsection on convexity of the balls.

It is recommended to first read this section, including the proof, to grasp the main ideas and gain broad understanding before proceeding to the next section.

For readers unfamiliar with convexity, we restate the definition of a convex set. The idea behind convexity is, that for every two points from a set, the line between these two points is also inside that set.

Definition 4.2.2 (Convexity)**.**
A set $A \subset \mathbb{R}^n$ is *convex*, iff $(1-\lambda)\mathbf{x} + \lambda\mathbf{y} \in A$ for all $\mathbf{x}, \mathbf{y} \in A$ and $\lambda \in [0, 1]$.

First, we begin with the statement of the theorem.

Theorem 4.2.3 (Homogeneity Theorem)**.**
Let S be an open convex subset of n-dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity (Eq. (2.2)) [7] *with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates* [8] *in S. If $\langle S, \delta\rangle$ is a metric space with additive segments, then δ is homogeneous, i.e., for any $\mathbf{x}, \mathbf{z} \in S$ and any real $0 \leq t \leq 1$,*

$$\delta[\mathbf{x}, \mathbf{x} + t(\mathbf{z} - \mathbf{x})] = t\,\delta(\mathbf{x}, \mathbf{z}).$$

Now, we continue with some remarks that make it possible to move the set such that $\mathbf{0} \in S$ by using the translation invariance of δ.

Remark 4.2.4 (Translation invariance and domain of δ and F). This remark establishes the continuity of δ and shows that we can assume without loss of generality, that $\mathbf{0} \in S$.

[7] i.e., δ takes the form $\delta\,(\mathbf{x}, \mathbf{y}) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$.

[8] Linear coordinates means that the vector representations are Cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the Cartesian product.

(i) δ is translation-invariant, i.e., $\delta(\mathbf{x}+\mathbf{y},\mathbf{z}+\mathbf{y}) = \delta(\mathbf{x},\mathbf{z})$:

$$\begin{aligned}\delta(\mathbf{x}+\mathbf{y},\mathbf{z}+\mathbf{y}) &= F[|x_1+y_1-(z_1+y_1)|,\ldots,x_n+|y_n-(z_n+y_n)|] \\ &= F[|x_1-z_1|,\ldots,|x_n-z_n|] \\ &= \delta(\mathbf{x},\mathbf{z})\end{aligned}$$

(ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{\mathbf{s}-\mathbf{x} : \mathbf{s} \in S\}$ for $\mathbf{x} \in S$ and $\delta'(\mathbf{a},\mathbf{b}) := \delta(\mathbf{a}+\mathbf{x},\mathbf{b}+\mathbf{x})$ for $\mathbf{a},\mathbf{b} \in S'$. Since $\mathbf{0} \in S'$, we also can introduce the following shorthand Notation $\delta'(\mathbf{x}') := \delta'(\mathbf{0},\mathbf{x}')$ for $\mathbf{x}' \in S'$. From now on, we assume that $\mathbf{0} \in S$, because by the argument just provided, we can shift S and δ such that $\mathbf{0} \in S$. Thus, we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta(t\cdot\mathbf{z}) = t\cdot\delta(\mathbf{z}) \qquad \forall t \in [0,1] \text{ and } \forall \mathbf{z} \in S'.$$

(iii) Since δ is defined by $\delta(\mathbf{x},\mathbf{y}) = F(|x_1-y_1|,\ldots,|x_n-y_n|)$ and δ is a metric, $F(\mathbf{x}) = 0$ iff $\mathbf{x} = 0$ must hold. Moreover, the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i-th argument. We denote this value as ρ_i and define it as

$$\rho_i := \sup\{|x_i-y_i| : (x_1,\ldots,x_n),(y_1,\ldots,y_n) \in S\} \in \mathbb{R}_{>0} \cup \{\infty\}.$$

Since δ is segmentally additive, every value between 0 and ρ_i must be in the domain of δ. Since this is true for all i, this implies that F is defined on a Cartesian product of half-open, real intervals of the form $[0,\rho_i)$.

(iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before proceeding to the proof, we first establish two preliminary statements that will be needed at the beginning. The first is intuitive and asserts that the distance between disjoint compact and closed sets is greater than zero.

Lemma 4.2.5.
If $K \subset \mathbb{R}^n$ is compact, $C \subset \mathbb{R}^n$ is closed and K and C are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty(\mathbf{p},\mathbf{p}') > \varepsilon$ for all $\mathbf{p} \in K$ and $\mathbf{p}' \in C$.

Proof. This statement follows directly from Thomson et al. (2008, p. 886, Exercise

13.12.10)[9], which asserts that there exist $\mathbf{p} \in K$ and $\mathbf{p}' \in C$ such that

$$\delta_\infty(\mathbf{p}, \mathbf{p}') = \inf\{\delta_\infty(\mathbf{x}, \mathbf{y}) : \mathbf{x} \in K, \mathbf{y} \in C\}.$$

If we assume the infimum to be zero, then by that theorem, $\delta_\infty(\mathbf{p}, \mathbf{p}') = 0$, which implies $\mathbf{p} = \mathbf{p}'$. This cannot be the case since K and C are disjoint. Therefore, the infimum must be strictly greater than zero. □

The second preliminary lemma allows us to make arbitrary small δ-balls which we will combine with the statement before to assure a distance in δ and not only in δ_∞.

Lemma 4.2.6 (Arbitrary small δ-balls)**.**
Let $\varepsilon > 0$ such that $\mathrm{B}^\infty_{(\varepsilon)} \subset S$. Then there exists $r > 0$ such that $\mathrm{B}^\delta_{(r)} \subset \mathrm{B}^\infty_{(\varepsilon)}$.

Proof. Since δ is a metric defined by $\delta(\mathbf{x}, \mathbf{y}) = F[|x_1 - y_1|, \ldots, |x_n - y_n|]$, we can define the j-th partial function of F by setting all other arguments to zero, i.e., $F^j(x_j) := F(0, \ldots, x_j, \ldots, 0)$. F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to Remark 4.2.4 (iii), $F^j(x_j) = 0$ if and only if $x_j = 0$ holds. Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $r_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(x_j) < r_j \implies x_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $\mathrm{B}^\infty_{(\varepsilon)} \subset S'$ implies that ε is a possible distance for each coordinate. (See also Remark 4.2.4 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \ldots n$.) Doing this for $j = 1, \ldots, n$ yields $r_1, \ldots, r_n$. By setting $r = \min\{r_1, \ldots, r_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(x_j) < r \leq r_j \implies x_j < \varepsilon \quad \forall j = 1, \ldots, n \implies \|\mathbf{x}\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta(\mathbf{x}) = F(\mathbf{x}) < r \implies F^j(x_j) < r \implies \|\mathbf{x}\|_\infty < \varepsilon.$$

□

[9] The complete statement of the theorem is: Given two closed subsets E and F of a metric space (X, d), where at least one of the sets is compact, then there exist points $e \in E$ and $f \in F$ such that

$$d(e, f) = \inf\{d(x, y) : x \in E, y \in F\}.$$

We now proceed to the proof of the homogeneity theorem. First, we restate the theorem in the form in which we will prove it.

Lemma (Restatement of the Homogeneity Theorem)**.**
Let S be an open convex subset of n-dimensional Euclidean space with $\mathbf{0} \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity (Eq. (2.2)) with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates in S. If $\langle S, \delta \rangle$ is a metric space with additive segments, then δ is homogeneous, i.e., for any $\mathbf{z} \in S$ and any real $0 \leq t \leq 1$,

$$\delta(t\mathbf{z}) = t\,\delta(\mathbf{z}).$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on the line from $\mathbf{0}$ to $\mathbf{z}$ are in S. Then we will prove the theorem for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment)**.**
Let $\mathbf{z} \in S$. Since S is open, $\mathbb{R}^n \setminus S$ is closed. The set $K := \{t\mathbf{z} : t \in [0,1]\}$ is compact, since continuous images($t \mapsto t\mathbf{z}$) of compact sets ($t \in [0,1]$) are compact. Thus, by Lemma 4.2.5 there exists a distance $\varepsilon > 0$ such that $\delta_\infty(\mathbf{p}, \mathbf{p}') < \varepsilon$ for all $\mathbf{p} \in K$ and $\mathbf{p}' \in \mathbb{R}^n \setminus S$. This means, for all points on the line from $\mathbf{0}$ to $\mathbf{z}$ the ball with radius ε is a subset of S, i.e., for all $t \in [0,1] : \mathrm{B}^\infty_\varepsilon(t\mathbf{z}) \subset S$. This is illustrated in Figure 4.1.

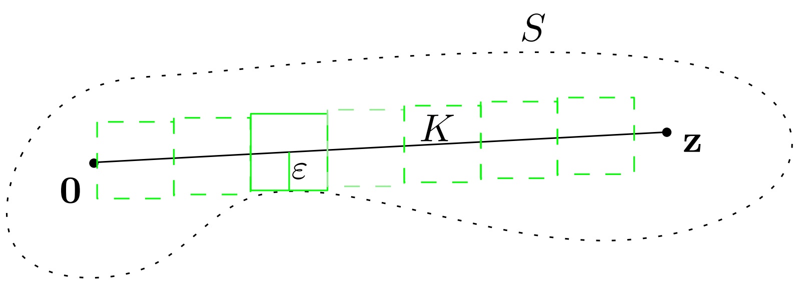

Figure 4.1: Illustration of the argument in Part 0.
The dotted black line represents the open set S and the black line represents the compact connection line between $\mathbf{0}$ and $\mathbf{z}$. The set $\mathrm{B}^\infty_{(\varepsilon)}$, represented as the green square, is a subset of S when moved along the line, as shown by the dashed green squares.

By Lemma 4.2.6, there exists $r > 0$ such that $\mathrm{B}^{\delta}_{(r)} \subset \mathrm{B}^{\infty}_{(\varepsilon)}$, and by translation invariance this holds for all points on the line: $\mathrm{B}^{\delta}_{r}(t\mathbf{z}) \subset S$ for all $t \in [0,1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\,\delta(\mathbf{z}) < r$. In particular, this means that $\mathrm{B}^{\delta}_{(r)}$ and therefore, $\mathrm{B}^{\delta}_{(\frac{1}{m}\,\delta(\mathbf{z}))}$ is bounded (with respect to the ∞ metric).

Part 1 ($t\,\delta(\mathbf{z}) \leq \delta(t\mathbf{z})$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Let $\mathbf{z^{(k)}} := \frac{k}{m} \cdot \mathbf{z}$ for $k = 0, \ldots, m$. Then $\mathbf{z^{(k)}} \in S$ for all k due to convexity of S. By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta\left(\mathbf{z^{(k-1)}}, \mathbf{z^{(k)}}\right) = \delta\left(\frac{k-1}{m}\mathbf{z}, \frac{k-1}{m}\mathbf{z} + \frac{1}{m}\mathbf{z}\right) = \delta\left(\frac{1}{m}\mathbf{z}\right) \quad \forall k = 1, \ldots, m.$$

Applying the triangle inequality $m-1$-times and using the above result yields

$$\delta(\mathbf{z}) \leq \sum_{k=0}^{m-1} \delta\left(\mathbf{z^{(k)}}, \mathbf{z^{(k+1)}}\right) = m \cdot \delta\left(\frac{1}{m}\mathbf{z}\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\,\delta(\mathbf{z}) \leq \delta\left(\frac{1}{m}\mathbf{z}\right)$, which concludes the first part.

Part 2 ($t\,\delta(\mathbf{z}) \geq \delta(t\mathbf{z})$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
The following arguments are illustrated in Figure 4.2.
Since δ is a metric with additive segments, there exists a segment from $\mathbf{0}$ to $\mathbf{z}$ with length $\delta(\mathbf{z})$. By Remark 3.1.2 the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\,\delta(\mathbf{z})$ and choose $\mathbf{y^{(0)}}, \ldots, \mathbf{y^{(m)}}$ equally spaced on the segment such that $\delta\left(\mathbf{y^{(k)}}, \mathbf{y^{(k-1)}}\right) = \frac{1}{m}\,\delta(\mathbf{z})$ and $\mathbf{y^{(0)}} = \mathbf{0}$ and $\mathbf{y^{(m)}} = \mathbf{z}$. Using this, we define $\mathbf{x^{(k)}} := \mathbf{y^{(k)}} - \mathbf{y^{(k-1)}}$. Translation invariance yields $\delta\left(\mathbf{x^{(k)}}\right) = \delta\left(\mathbf{y^{(k)}}, \mathbf{y^{(k-1)}}\right) = \frac{1}{m}\,\delta(\mathbf{z})$ and, due to the choice of m in Part 0, $\mathbf{x^{(k)}} \in S$ holds. Note, that this is still true, if we increase m.

Further, we can observe that $\frac{1}{m}\mathbf{z}$ is a convex combination of $\mathbf{x^{(k)}}$:

$$\sum_{k=1}^{m} \frac{1}{m}\mathbf{x^{(k)}} = \frac{1}{m}(\mathbf{y^{(1)}} - \mathbf{y^{(0)}} + \mathbf{y^{(2)}} - \mathbf{y^{(1)}} + \cdots + \mathbf{y^{(m)}}) = \frac{1}{m}(-\mathbf{y^{(0)}} + \mathbf{y^{(m)}}) = \frac{1}{m}\mathbf{z}.$$

We showed before, that $\delta\left(\mathbf{x^{(k)}}\right) = \frac{1}{m}\,\delta(\mathbf{z})$, which implies $\mathbf{x^{(k)}} \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(\mathbf{z})]}$. By Part 0 $\mathrm{B}^{\delta}_{(\frac{1}{m}\,\delta(\mathbf{z}))}$ is bounded. Therefore, we can apply Lemma 4.2.7 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}\mathbf{z} \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(\mathbf{z})]}$ must hold. Thus, $\delta\left(\frac{1}{m}\mathbf{z}\right) \leq \frac{1}{m}\,\delta(\mathbf{z})$ which concludes the second part of the proof.

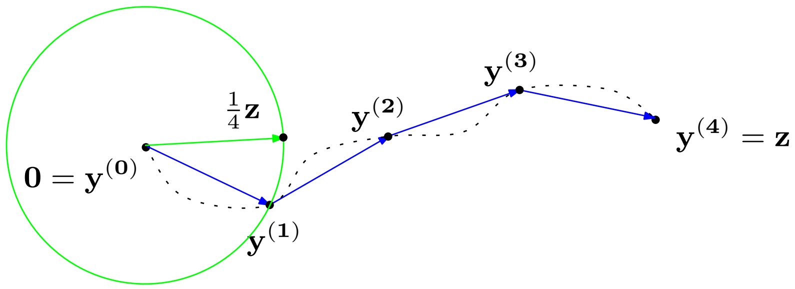

Figure 4.2: Illustration of the argument in Part 2 for $m = 4$.
The dotted black line is the segment from $\mathbf{0}$ to $\mathbf{z}$. The black points represent the equally spaced $\mathbf{y^{(k)}}$ on the line, whereas the blue vectors are the $\mathbf{x^{(k)}}$. We can see that the convex combination of the blue vectors yields the green vector which is inside the given sphere, if balls are convex. Note, that the sphere is intuitively illustrated as a circle; however this is generally not the case.

Now, by combining Part 1 and Part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in Part 2 for $\mathbf{x^{(k)}}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in Part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \ldots, m$.

Part 3 ($t\,\delta(\mathbf{z}) = \delta(t\mathbf{z})$ for $t = \frac{k}{m}$ and $k \in \{0, \ldots, m\}$)**.**
For $\mathbf{z^{(1)}}$ we already have $\delta\left(\mathbf{0}, \mathbf{z^{(1)}}\right) = \frac{1}{m}\delta(\mathbf{0}, \mathbf{z})$. Now, for consecutive points, by translation invariance, it holds that

$$\delta\left(\mathbf{z^{(k-1)}}, \mathbf{z^{(k)}}\right) = \delta\left(\mathbf{z^{(k-2)}}, \mathbf{z^{(k-1)}}\right) = \cdots = \delta\left(\mathbf{z^{(1)}}, \mathbf{z^{(0)}}\right) = \delta\left(\mathbf{z^{(1)}}, \mathbf{0}\right) = \frac{1}{m}\delta(\mathbf{0}, \mathbf{z}).$$

Using this and the triangle inequality, we get

$$\begin{aligned}\delta(\mathbf{0}, \mathbf{z}) &\leq \delta\left(\mathbf{0}, \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}}, \mathbf{z^{(2)}}\right) + \cdots + \delta\left(\mathbf{z^{(m-1)}}, \mathbf{z^{(m)}}\right)\\ &= n \cdot \frac{1}{n}\delta(\mathbf{0}, \mathbf{z})\\ &= \delta(\mathbf{0}, \mathbf{z}).\end{aligned}$$

This implies

$$\delta(\mathbf{0}, \mathbf{z}) = \delta\left(\mathbf{0}, \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}}, \mathbf{z^{(2)}}\right) + \cdots + \delta\left(\mathbf{z^{(m-1)}}, \mathbf{z^{(m)}}\right). \tag{4.8}$$

For the distance $\delta\left(\mathbf{0}, \mathbf{z^{(2)}}\right)$, we get the following two inequalities.
Firstly,

$$\delta\left(\mathbf{0}, \mathbf{z^{(2)}}\right) \leq \delta\left(\mathbf{0}, \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}}, \mathbf{z^{(2)}}\right) = \frac{2}{m}\delta(\mathbf{0}, \mathbf{z}).$$

Secondly,

$$\begin{aligned}
\delta\left(\mathbf{0},\mathbf{z}\right) &\leq \delta\left(\mathbf{0},\mathbf{z}^{(\mathbf{2})}\right) + \delta\left(\mathbf{z}^{(\mathbf{2})},\mathbf{z}\right) \\
\delta\left(\mathbf{0},\mathbf{z}^{(\mathbf{2})}\right) &\geq \delta\left(\mathbf{0},\mathbf{z}\right) - \underbrace{\delta\left(\mathbf{z}^{(\mathbf{2})},\mathbf{z}\right)}_{\leq\sum_k \delta\left(\mathbf{z}^{(\mathbf{2+k})},\mathbf{z}^{(\mathbf{2+k+1})}\right)} && \text{(Rearrange)} \\
&\geq \delta\left(\mathbf{0},\mathbf{z}\right) - \sum_{k=0}^{m-3} \delta\left(\mathbf{z}^{(\mathbf{2+k})},\mathbf{z}^{(\mathbf{2+k+1})}\right) && (\triangle \text{ inequality}) \\
&= \delta\left(\mathbf{0},\mathbf{z}\right) - \frac{m-2}{m}\,\delta\left(\mathbf{0},\mathbf{z}\right) && \text{(using Eq. (4.8))} \\
&= \frac{2}{m}\,\delta\left(\mathbf{0},\mathbf{z}\right).
\end{aligned}$$

Combining these two inequalities yields $\delta\left(\mathbf{0},\mathbf{z}^{(\mathbf{2})}\right) = \frac{2}{m}\,\delta\left(\mathbf{0},\mathbf{z}\right)$. By repeating this for $\delta\left(\mathbf{0},\mathbf{z}^{(\mathbf{k})}\right)$ for every $k = 2,\ldots,m-1$, we get the equality for every rational number for m big enough. However, every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m. This yields, that the result is true for every rational t.

Part 4 ($t\,\delta\left(\mathbf{z}\right) = \delta\left(t\mathbf{z}\right)$ for $t \in [0,1]$)**.**
The final step can be completed using continuity of the metric. Now let $t \in [0,1]$ and $t_n \in \mathbb{Q}$ with $t_n \to t$ for $n \to \infty$. Then

$$\begin{aligned}
\delta\left(t\mathbf{z}\right) &= \delta\left(\lim_{n\to\infty} t_n\mathbf{z}\right) \\
&= \lim_{n\to\infty} \delta\left(t_n\mathbf{z}\right) && \text{(continuity)} \\
&= \lim_{n\to\infty} t_n\,\delta\left(\mathbf{z}\right) && (t_n \in \mathbb{Q} \text{ and Part 3}) \\
&= t\,\delta\left(\mathbf{z}\right).
\end{aligned}$$

□

Convexity of δ Balls

In this subsection we will prove the following lemma which is needed to conclude the proof of Theorem 4.2.3. Everything will be assumed as in Theorem 4.2.3. Note, that all topological terms are with respect to the natural topology of the Euclidean space.

Lemma 4.2.7 (Convexity of δ-Spheres)**.**
For $r > 0$ if $\mathrm{B}^{\delta}_{[r]} \subset S$ is bounded, then it is convex.

First, we will state some definitions and results on basic convex geometry that we need, and prove the lemma at the end of this section. The definitions and results on convex geometry are taken out of Hug and Weil (2020).

We begin by introducing the notions of convex combinations, the convex hull and extreme points. Recall, that a set is convex if for all two pairs of points of that set, the linear interpolation between these points stays inside the set. By increasing the amount of points, we get what is called the convex combination of these points. A convex combination is similar to a linear combination, but has the additional constraint that the coefficients must be positive and add up to 1. The convex hull of a set is, like any other hull, the smallest convex set that contains the set.

Definition 4.2.8 (Convex Combination and Hull).

1. Let $k \in \mathbb{N}$, let $\mathbf{x_1}, \dots, \mathbf{x_k} \in \mathbb{R}^n$, and let $\lambda_1, \dots, \lambda_k \in [0,1]$ be such that $\lambda_1 + \dots + \lambda_k = 1$. Then $\lambda_1 \mathbf{x_1} + \dots + \lambda_k \mathbf{x_k}$ is called a *convex combination* of the points $\mathbf{x_1}, \dots, \mathbf{x_k}$.

2. For $A \subset \mathbb{R}^n$ the *convex Hull* $\operatorname{conv}(A)$ is the smallest convex set containing A, i.e.,
$$\operatorname{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 4.2.9.
For $A \subset \mathbb{R}^n$ the convex Hull is the set of all convex combination of points in A:
$$\operatorname{conv}(A) = \left\{ \sum_{i=1}^{k} \lambda_i \mathbf{x_i} : k \in \mathbb{N}, \mathbf{x_1}, \dots, \mathbf{x_k} \in A, \lambda_1, \dots, \lambda_k \in [0,1] : \sum_{i=1}^{k} \lambda_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 4.2.10 (convex Hull of compact set is compact).
If $A \subset \mathbb{R}^n$ is compact, then $\operatorname{conv}(A)$ *is compact.*

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 4.2.11 (Extreme Point).
Let $A \subset \mathbb{R}^n$ be closed and convex. A point $\mathbf{x} \in A$ is called an *extreme point of A* if $\mathbf{x}$ cannot be represented as a nontrivial (or strict) convex combination of points of A, that is, if $\mathbf{x} = \lambda \cdot \mathbf{y} + (1 - \lambda) \cdot \mathbf{z}$ with $\mathbf{y}, \mathbf{z} \in A$ and $\lambda \in (0, 1)$, implies that $\mathbf{x} = \mathbf{y} = \mathbf{z}$. The set of all extreme points of A is denoted by extreme(A).

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 4.2.12 (Minkowski Theorem).
Let $A \subset \mathbb{R}^n$ be compact and convex. Then $A = \mathrm{conv}(\mathrm{extreme}\, A)$.

Before proving the convexity of δ-balls, we will first establish two lemmas that will be needed in the proof. The first one is the openness/closedness of δ-balls.

Lemma 4.2.13 (δ-Balls are open/closed).
For every radius $r > 0$, the δ-Balls are open $(\mathrm{B}^{\delta}_{(r)})$ *or closed* $(\mathrm{B}^{\delta}_{[r]})$.

Proof. $\delta : S' \times S' \to \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $\mathbf{y} \mapsto \delta(\mathbf{y}) = \delta(\mathbf{0}, \mathbf{y})$ is continuous. Consider the closed interval $U' = [0, r] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned} \delta^{-1}(U') &= \{\mathbf{y} \in S' : \delta(\mathbf{y}) \in U'\} && \text{(Preimage definition)} \\ &= \{\mathbf{y} \in S' : \delta(\mathbf{0}, \mathbf{y}) \leq r\} && \text{(Definition of } \delta) \\ &= \mathrm{B}^{\delta}_{[r]} . \end{aligned}$$

Due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $\mathrm{B}^{\delta}_{[r]}$ is closed. By translation invariance this holds for closed balls with any center in S. Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. □

The second lemma shows, that extreme points of balls are compressed by a factor of $\frac{1}{m} \in \mathbb{N}$, if the radius of the ball gets reduced by the same factor.

Lemma 4.2.14 (compressing extreme points).
Let $r \in \mathbb{R}_{>0}$ and $m \in \mathbb{N}_{>0}$. Further let $\mathrm{E}(r) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ *and* $\mathrm{C}[r] := \mathrm{conv}(\mathrm{B}^{\delta}_{[r]})$. *Then,* $\mathbf{y} \in \mathrm{E}(r)$ *implies* $\frac{1}{m}\mathbf{y} \in \mathrm{E}\left(\frac{r}{m}\right)$.

Remark. We use square brackets for $\mathrm{C}[r]$ to denote that this set is closed. For the reasoning, see the first argument in Section 4.2.1.

Proof. We prove this statement by contraposition.

Assume, that $\frac{1}{m}\mathbf{y}$ is not an extreme point, i.e., $\frac{1}{m}\mathbf{y} \in \mathrm{C}\left[\frac{r}{m}\right] \setminus \mathrm{E}\left(\frac{r}{m}\right)$. Then we can write $\frac{1}{m}\mathbf{y}$ as a strict convex combination of elements $\mathbf{x^{(k)}} \in \mathrm{C}\left[\frac{r}{m}\right]$:

$$\frac{1}{m}\mathbf{y} = \sum_{i=1}^{p} \lambda_i \mathbf{x^{(k)}} \quad \text{with} \sum_{i=1}^{p} \lambda_i = 1, \quad \lambda_i > 0,\ \mathbf{x^{(k)}} \in \mathrm{C}\left[\frac{r}{m}\right] \text{ and } p \in \mathbb{N}$$

But because $\mathrm{C}\left[\frac{r}{m}\right] = \mathrm{conv}(\mathrm{B}^{\delta}_{[\frac{r}{m}]})$ holds, we can write any element from $\mathrm{C}\left[\frac{r}{m}\right]$ as a convex combination of elements from $\mathrm{B}^{\delta}_{[\frac{r}{m}]}$, and therefore we can assume $\mathbf{x^{(k)}} \in \mathrm{B}^{\delta}_{[\frac{r}{m}]}$. By translation invariance and the triangle inequality we get for all i:

$$\begin{aligned}
\delta\left(m\mathbf{x^{(k)}}\right) &\le \delta\left(0, \mathbf{x^{(k)}}\right) + \delta\left(\mathbf{x^{(k)}}, m\mathbf{x^{(k)}}\right) && \text{triangle inequality}\\
&= \delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0, (m-1)\cdot\mathbf{x^{(k)}}\right) && \text{translation invariance}\\
&\le \delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0, \mathbf{x^{(k)}}\right) + \delta\left(\mathbf{x^{(k)}}, (m-1)\cdot\mathbf{x^{(k)}}\right) && \text{triangle inequality}\\
&= 2\cdot\delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0, (m-2)\cdot\mathbf{x^{(k)}}\right) && \text{translation invariance}\\
&\dots\\
&\le m\cdot\delta\left(\mathbf{x^{(k)}}\right)\\
&= m\cdot\frac{r}{m} = r\\
\implies m\mathbf{x^{(k)}} &\in \mathrm{B}^{\delta}_{[r]}\,.
\end{aligned}$$

This yields, that $\mathbf{y} = \sum_{i=1}^{p} \lambda_i m\mathbf{x^{(k)}}$ is a strict convex combination (because $\lambda_i m \neq 0$) and therefore $\mathbf{y}$ is not in $\mathrm{E}(r)$. □

Now we can prove the convexity of δ-balls.

Proof of Lemma 4.2.7. Let $\mathrm{E}(r) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ and $\mathrm{C}[r] := \mathrm{conv}(\mathrm{B}^{\delta}_{[r]})$. By Lemma 4.2.13 $\mathrm{B}^{\delta}_{[r]}$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $\mathrm{B}^{\delta}_{[r]}$ is compact. By definition and Theorem 4.2.10 it follows, that $\mathrm{C}[r]$ is compact (and therefore closed, which is the reason for our square bracket notation) and convex. Application of the Minkowski Theorem (Theorem 4.2.12) yields

$$\mathrm{C}[r] = \mathrm{conv}(\mathrm{extreme}(\mathrm{C}[r])).$$

Further, it holds that $\mathrm{extreme}(\mathrm{C}[r]) = \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$\mathrm{B}^{\delta}_{[r]} \subset \mathrm{C}[r] = \mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})).$$

Now, if we prove that $\mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})) \subset \mathrm{B}^{\delta}_{[r]}$, we get that $\mathrm{B}^{\delta}_{[r]} = \mathrm{C}\,[r]$ which proves the convexity of $\mathrm{B}^{\delta}_{[r]}$. We simplify this statement before proving it.

$$\mathrm{conv}(\underbrace{\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})}_{\mathrm{E}(r)}) \subset B$$

$$\Updownarrow$$

$$\sum_{i=1}^{k} \lambda_i \mathbf{y}^{(\mathbf{i})} \in \mathrm{B}^{\delta}_{[r]}, \quad k \in \mathbb{N}_{>0}, \begin{cases} \mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\,(r) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^{k} \lambda_i = 1 \end{cases} \qquad \text{(Definition)}$$

$$\Uparrow$$

$$\sum_{i=1}^{k} \lambda_i \mathbf{y}^{(\mathbf{i})} \in \mathrm{B}^{\delta}_{[r]}, \quad k \in \mathbb{N}_{>0}, \begin{cases} \mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\,(r) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^{k} \lambda_i = 1 \end{cases} \qquad (B \text{ closed, } \mathbb{Q} \subset \mathbb{R} \text{ dense } ^{10})$$

$$\Uparrow$$

$$\sum_{i=1}^{m} \frac{1}{m} \mathbf{y}^{(\mathbf{i})} \in \mathrm{B}^{\delta}_{[r]}, \quad m \in \mathbb{N}_{>0}, \; \mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\,(r) \qquad (m \text{ common denominator})$$

Thus, we will continue proving that for $m \in \mathbb{N}_{>0}$ and $\mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\,(r)$ it holds that $\sum_{i=1}^{m} \frac{1}{m} \mathbf{y}^{(\mathbf{i})} \in \mathrm{B}^{\delta}_{[r]}$.

For $\mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\,(r)$ it holds by Lemma 4.2.14 that $\frac{1}{m}\mathbf{y}^{(\mathbf{i})} \in \mathrm{E}\left(\frac{r}{m}\right) \subset \mathrm{B}^{\delta}_{[\frac{r}{m}]}$. This means, that for the metric distance of the convex combination the following holds:

$$\begin{aligned}
\delta\left(\mathbf{0}, \sum_{i=1}^{m} \frac{1}{m} \mathbf{y}^{(\mathbf{i})}\right) &\leq \delta\left(\mathbf{0}, \frac{1}{m}\mathbf{y}^{(\mathbf{1})}\right) + \delta\left(\frac{1}{m}\mathbf{y}^{(\mathbf{1})}, \sum_{i=1}^{m} \frac{1}{m}\mathbf{y}^{(\mathbf{i})}\right) && \text{(triangle inequality)} \\
&= \delta\left(\mathbf{0}, \frac{1}{m}\mathbf{y}^{(\mathbf{1})}\right) + \delta\left(\mathbf{0}, \sum_{i=2}^{m} \frac{1}{m}\mathbf{y}^{(\mathbf{i})}\right) && \text{(translation invariance)} \\
&\leq \frac{r}{m} + \delta\left(\mathbf{0}, \sum_{i=2}^{m} \frac{1}{m}\mathbf{y}^{(\mathbf{i})}\right) && \text{(translation invariance)} \\
&\dots && \text{(repeat)} \\
&\leq m\frac{r}{m} = r
\end{aligned}$$

This yields that $\sum_{i=1}^{m} \frac{1}{m} \mathbf{y}^{(\mathbf{i})} \in \mathrm{B}^{\delta}_{[r]}$ holds which proves the statement. □

4.2.2 Uniqueness Theorem for Homogeneous Means

In this section we will prove the uniqueness theorem for homogeneous means, which states that the power means [11] (including the geometric mean as a limit case) are the only homogeneous quasi-arithmetic means. Before beginning the proof, we will first define quasi-arithmetic means and one class of this means, the power means. Both, the proof and the definitions are based on Hardy et al. (1952) and Bullen (2003).

We begin with defining quasi-arithmetic means. The name stems from generalizing the arithmetic mean by applying a function on each value before averaging and then using its inverse to transform the result back.

Definition 4.2.15 (Quasi Arithmetic Means and Homogeneity)**.**
Let $\phi : [0, \infty) \to [0, \infty)$ be continuous and strictly increasing and let $\mathbf{x}, \mathbf{w} \in \mathbb{R}^n_{\geq 0}$. Then the *quasi-arithmetic ϕ-mean of* $\mathbf{x}$ *with weights* $\mathbf{w}$ is defined as

$$\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i \phi(x_i)\right).$$

The mean is said to be *homogeneous*, if

$$\mathcal{M}_\phi(t \cdot \mathbf{x},\ \mathbf{w}) = t \cdot \mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$$

holds for all $t > 0$.

Next, we define the power means and remark that they are quasi-arithmetic means.

Definition 4.2.16 (Power mean)**.**
For $n \in \mathbb{N}, \mathbf{x}, \mathbf{w} \in \mathbb{R}^n_{\geq 0}$ and $p > 0$ we define the *power means* $\mathcal{M}_p(\mathbf{x}, \mathbf{w})$ as

$$\mathcal{M}_p(\mathbf{x}, \mathbf{w}) = \left(\frac{1}{\sum_{i=1}^{n} w_i} \sum_{i=1}^{n} w_i x_i^p\right)^{1/p}.$$

For $p = 0$ we can define $\mathcal{M}_0(\mathbf{x}, \mathbf{w})$ as the *geometric mean*

$$\mathcal{M}_0(\mathbf{x}, \mathbf{w}) = \left(\prod_{i=1}^{n} x_i^{w_i}\right)^{1/\sum_{i=1}^{n} w_i}$$

since it is obtained as a limit case for $r \to 0$ (Bullen, 2003, p. 176).

[11] In some literature this is also referred to as the Hölder mean or the generalized means.

Remark 4.2.17 (Power Means are Quasi-Arithmetic)*.* By setting $\phi(x) = x^p$ for $p > 0$ or $\phi(x) = \log(x)$ we obtain the power means for $p > 0$ and $p = 0$ respectively. In particular, setting $w_i = 1$ yields the familiar generalizations of the power mean

$$\left(\frac{1}{n}\sum_{i=1}^{n} x_i^p\right)^{1/p}$$

and the arithmetic mean

$$\sqrt[n]{\prod_{i=1}^{n} x_i}.$$

Now we can continue with a theorem about equal means, which is needed in the proof of the uniqueness theorem for homogeneous means.

Theorem 4.2.18 (Characterization of Equal Means)**.**
In order that

$$\mathcal{M}_\chi(\mathbf{a},\ \mathbf{w}) = \mathcal{M}_\psi(\mathbf{a},\ \mathbf{w}) \tag{4.9}$$

for all $\mathbf{a}, \mathbf{w}$ *and* $n \in \mathbb{N}$*, it is necessary and sufficient that*

$$\phi = \alpha \cdot \psi + \beta \tag{4.10}$$

where α *and* β *are constants and* $\alpha \neq 0$*.*

Proof.

Part 1 ($\Rightarrow$)**.**
Assume that Equation (4.9) holds. In particular, it also holds for $n = 2$. Thus, we have

$$\chi^{-1}(w_1\chi(a_1) + w_2\chi(a_2)) = \psi^{-1}(w_1\psi(a_1) + w_2\psi(a_2))$$

Putting $\phi(x) = \chi(\psi^{-1}(x)), x_i = \psi(a_i)$ for $i = 1, 2$ yields

$$\phi(w_1x_1 + w_2x_2) = w_1\psi(x_1) + w_2\psi(x_2) \quad x_1, x_2 \in \mathbb{R}_{\geq 0}.$$

In particular, this holds for $w_1, w_2 = 1/2$ which then reduces to

$$\phi\left(\frac{x_1 + x_2}{2}\right) = \frac{\phi(x_1)}{2} + \frac{\phi(x_2)}{2} \quad x_1, x_2 \in \mathbb{R}_{\geq 0}.$$

By Aczél (1966, p. 46), the general continuous solution is $\psi(x) = \alpha x + \beta$, hence $\chi = \alpha\psi + \beta$.

Part 2 ($\Leftarrow$).
Assume that Equation (4.10) holds. Then

$$\begin{aligned} \chi(\mathcal{M}_\chi(\mathbf{a},\ \mathbf{w})) &= \sum_{i=1}^{n} w_i \chi(a_i) && \text{(cancellation of inverses)} \\ &= \sum_{i=1}^{n} q_i \alpha\psi(a_i) + \beta && \text{(Eq. (4.10))} \\ &= \alpha\left(\sum_{i=1}^{n} w_i\psi(a_i)\right) + \beta && \text{(rearranging)} \\ &= \chi\left[\mathcal{M}_\psi(\mathbf{a},\ \mathbf{w})\right] && \text{(Eq. (4.10))} \end{aligned}$$

Applying χ^{-1} to both sides of the equation yields Equation (4.9).

□

Now we finally state and prove the main theorem of this section.

Theorem 4.2.19 ($\mathcal{M}_p$ are the only homogeneous means $\mathcal{M}_\phi$)**.**
Suppose that ϕ is continuous and strictly increasing in the open interval $(0, \infty)$ and that the mean defined by ϕ is homogeneous, i.e.,

$$\mathcal{M}_\phi(t\mathbf{a},\ \mathbf{w}) = t\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w}) \tag{4.11}$$

for all positive $\mathbf{a}, \mathbf{w}, t$ and all $n \in \mathbb{N}$ such that $\sum_{i=1}^{n} w_i = 1$. Then $\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w})$ is $\mathcal{M}_p(\mathbf{a}, \mathbf{w})$.

Proof. By Theorem 4.2.18 we can assume that $\phi(1) = 0$, since we can replace $\phi(x)$ by $\phi(x) - \phi(1)$ without changing $\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w})$. Next, we write Equation (4.11) as

$$\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w}) = t^{-1}\mathcal{M}_\phi(t\mathbf{a},\ \mathbf{w}) = t^{-1}\phi^{-1}\left(\sum_{i=1}^{n} w_i\phi(ta_i)\right) = \mathcal{M}_\psi(\mathbf{a},\ \mathbf{w})$$

by setting $\psi(x) = \phi(tx)$. From this, it follows that $\psi^{-1}(x) = t^{-1} \cdot \phi(x)$[12], which in turn justifies the above equation. Therefore, we have the equality of the means defined by ψ and ϕ. Since ψ is defined for every $t > 0$, we have by Theorem 4.2.18 that there are $\alpha(t), \beta(t) \in \mathbb{R}, \alpha(t) \neq 0$ such that

$$\phi(tx) = \psi(x) = \alpha(t) \cdot \phi(x) + \beta(t).$$

[12]For this, let $\psi^{-1}(x) = y$. Then, by applying ψ to both sides, and using its definition, this yields $x = \phi(ty)$. Applying the inverse of ϕ and dividing by t yields $t^{-1}\phi^{-1}(x) = y$, which in turn is equal to $\psi^{-1}(x)$ by the first equation.

Plugging in $x = 1$ and using $\phi(1) = \phi(0)$ yields

$$\phi(t) = \psi(1) = \beta(t).$$

Thus, for all positive x, y by setting $y = t$

$$\phi(xy) = \alpha(y)\phi(x) + \phi(y) \tag{4.12}$$

Similarly, by exchanging the roles of x and y we get

$$\phi(xy) = \alpha(x)\phi(y) + \phi(x)$$

Combining these two equalities yields

$$\begin{aligned}
\alpha(y)\phi(x) + \phi(y) &= \alpha(x)\phi(y) + \phi(x) && (-\phi(x) - \phi(y)) \\
\alpha(y)\phi(x) - \phi(x) &= \alpha(x)\phi(y) - \phi(y) && (\text{divide }^{13}\text{by } \phi(x) \cdot \phi(y)) \\
\frac{\alpha(x) - 1}{\phi(x)} &= \frac{\alpha(y) - 1}{\phi(y)} := c
\end{aligned}$$

Since this must hold for all x and y, the expressions must reduce to a constant c. This can be seen by settings, for example, $x = 1$ and observing that the right-hand side must equal to that for all y and therefore must be constant. The same argument can be applied in reverse. Using this, we get the following expression

$$\alpha(y) = 1 + c\phi(y).$$

Plugging this expression for $\alpha(y)$ into Equation (4.12) yields

$$\phi(xy) = c\phi(x)\phi(y) + \phi(x) + \phi(y). \tag{4.13}$$

For this equation, there are two cases to consider: $c = 0$ and $c \neq 0$. If $c = 0$, then Equation (4.13) reduces to the well known logarithmic Cauchy equation

$$\phi(xy) = \phi(x) + \phi(y),$$

of which the most general and for $x > 0$ continuous solution is $\phi(x) = \alpha \log(x)$ for some constant $\alpha \in \mathbb{R}$ (Aczél, 1966, p. 39; Cauchy, 1821, p. 110-111).

If $c \neq 0$, we set $f(x) = c\phi(x)+1$ and Equation (4.13) reduces to the multiplicative Cauchy functional equation

$$f(xy) = f(x)f(y),$$

whose general solution is $f(x) = x^r$ (Aczél, 1966, p. 39; Cauchy, 1821, p. 111-112). Substituting back yields $\phi(x) = \frac{x^r-1}{c}$. □

[13] Provided $x \neq 1, y \neq 1$. Since Equation (4.13) is true when x or y is 1, we can ignore the exception.

Sources and Original Contributions

The main results of this chapter, namely Theorem 4.1.1 and Theorem 4.2.1, were first published in Beals et al. (1968), then proved in Tversky and Krantz (1970) and finally refined and updated to the version presented in Krantz et al. (1971, Chapter 14.4.5, pp. 185-187). This chapter is primarily based on the latter.
The proofs presented in Section 4.1 extend some arguments from Krantz et al. (1971, pp. 192-194) and include additional references.

The proofs presented in Section 4.2 largely extend and complete the proofs provided in Krantz et al. (1971, pp. 195-199) and Tversky and Krantz (1970, pp. 589-593)[14]. In both sources, the statements and proofs are almost identical. The proof of the main theorem, Theorem 4.2.1, can mostly be found in both sources. However, important details are missing in this proof. Most notably, the preconditions for the application of Theorem 4.2.19 have neither been mentioned nor established.

The proofs of the homogeneity theorem (Section 4.2.1) and the convexity of δ balls (Section 4.2.1) are briefly outlined; however, some arguments are either skipped or only briefly mentioned without further explanation. Moreover, the statement of the homogeneity theorem is incorrect, as continuity is used but not assumed. Part 1 and Part 2 of the proof can be found in both references. The other parts of the proof are briefly outlined, so the majority of the work in this section involves the author completing the proofs. The definitions and convexity results in Section 4.2.1 are based on Chapters 1.1 to 1.4 of Hug and Weil (2020, pp. 1-38). Most other arguments have been briefly outlined.

The section on homogeneous quasi-arithmetic means, Section 4.2.2, is based on the work found in Hardy et al. (1952, pp. 66-69) and Bullen (2003, pp. 271-273). The proofs provided here can be found in these references.

[14]Until the end of this section, all references are with respect to these two sources, e.g., when writing “the proofs are briefly outlined”

Chapter 5

Discussion

Now we briefly discuss the results from a psychological perspective. In the AD model (Eq. (2.4)) the functions F_i and φ can be interpreted as two distinct processes that lead to similarity judgements: a perceptual process that satisfies intradimensional subtractivity and an evaluative process that satisfies interdimensional additivity (Tversky & Krantz, 1970). The perceptual process is represented by φ, known as psychophysical functions, while the evaluative process is captures by F_i, referred to as similarity functions. Furthermore, Tversky and Krantz (1970) argue that interdimensional additivity and intradimensional subtractivity can be considered as defining properties for psychological dimensions in the context of multidimensional similarity. The outcomes outlined in this thesis provide methods for testing whether a factor satisfies these properties and, therefore, is a candidate for a psychological dimension.

However, arguably, the presented results have the greatest implications for the theory of multidimensional scaling (MDS) which is a widely used data analysis technique (for example usages of MDS in psychology see Jaworska and Chupetlovska-Anastasova (2009)). The goal of (ordinal) multidimensional scaling is to embed the stimuli objects in some multidimensional metric space (usually Euclidean) in such a way that the dissimilarity orderings are represented, as accurately as possible, by the metric distance between the points. Thus, objects that are more similar are placed closer together, and the dimensionality of the metric space is typically chosen to be as low as possible while still maintaining an accurate representation. In most contexts, MDS is viewed as a method for data reduction and is often used without proper justification, as its underlying theory has not been explored much (Beals et al., 1968). The embedding of ordinal stimuli in a metric space is not trivial, and these results illustrate the ordinal assumptions that must be made to justify such an embedding. Therefore, the results presented in this thesis can be interpreted in two ways: first, as conditions that must be tested to justify the use of MDS, and, second, as psychological theories about mental representations that can be applied to study psychological similarity (Beals et al., 1968; Krantz et al., 1971; Tversky & Krantz, 1970).

The idea of metric and dimensional representations are intuitively and computationally very appealing. Note, that these assumptions are, distinct: the results in Chapter 3 do not rely on a dimensional structure, while Chapter 2 does not assume a metric representation. However, some of their fundamental assumptions present difficulties in justifying their use. These, along with some axioms, will be discussed next.

In representational measurement, in addition to distinguishing between necessary and sufficient axioms, there is also a distinction between testable and non-testable axioms. Non-testable (or technical) axioms cannot be empirically falsified, whereas testable axioms can. The axioms of a complete segmentally additive proximity structure (Definition 3.2.3) are both necessary and sufficient (Krantz et al., 1971, p. 168). In the AD structure (Definition 2.2.6), all axioms except restricted solvability are necessary (Krantz et al., 1971, p. 184).
In the following we discuss some testable and non-testable axioms. First, we discuss the Archimedean Axiom, which is a fundamental and technical axiom in representational measurement. Its name stems from the Archimedean property of the real numbers, which states that for every pair of real numbers $x < y$ there exists an integer n such that $nx > y$. This can be reinterpreted as: given the sequence $x, x+x, x+x+x\ldots$ then the number of terms smaller than y is bounded. Interpreting this property in this way reveals its close connection to both the Archimedean Axiom for AD structures (Axiom 2.2.4) and for segmentally additive proximity structures (Definition 3.2.3). In both cases, we require a finite sequence of equally spaced[1] objects that exceeds or reaches some point or radius. As we have seen in Section 3.2, this axiom is fundamental for the construction of distance between objects. For further discussion see Krantz et al. (1989, pp. 25-26) and Krantz et al. (1971, pp. 165-167).

Now, we proceed with the discussion of testable axioms. Two of the basic assumptions of proximity structures are symmetry and that self-similarity is strict greater than other-similarity. However, these are not always satisfied. In the case

[1]We do require the spacing to be exactly equal (as in the Archimedean property for real numbers), but rather that it is bounded.

where two stimuli are physically different but indistinguishable to the observer, then self-similarity is not strictly greater than other-similarity. In this case, one can instead conduct the analysis on the equivalence classes if they are well-defined (Krantz et al., 1971, p. 161). Regarding symmetry, this does not hold true in many cases, which has been demonstrated for perceptive and conceptual comparisons. For example, an ellipse is more similar to a circle than a circle to an ellipse and North Korea is more similar to China than China is to North Korea (Tversky, 1977; Tversky & Gati, 1979). For general metric models, testing the triangle comes with challenges, as it requires numerical addition, which is not possible with ordinal data. Krantz et al. (1971, pp. 205-207) present a method to test it under the assumptions of segmental additivity and a dimensional structure. Furthermore, the essence of the metric model lies in segmental solvability and testing this axiom is fairly difficult. It requires constructing multiple isosimilarity contour by obtaining multiple solutions multiple equation. Then a study of the similarity between item on the contours has to be conducted (Beals et al., 1968; Krantz et al., 1971, p. 165). In the case of AD Structures, decomposability, (interdimensional) additivity and (intradimensional) subtractivity have been tested (Krantz & Tversky, 1975; Tversky & Krantz, 1969; Wender, 1971). Since both, additivity and subtractivity imply one-factor independence, this property was used to test an ordinal version of translation invariance. By one-factor independence, changing a common value on one factor does not change the similarity ordering. This corresponds to a translation from the initial common value to the new common value after the change. This was used in a study by Gati and Tversky (1984), which demonstrated highly predictable violations of this property. In this study, participants rated the dissimilarity between pairs of schematic faces (hand-drawn faces) that varied along three dimensions: the presence or absence of eyeglasses, a hat and a beard. The systematic violation stems from the fact, that, in this study, adding a common attribute to a pair of faces – where the attribute was previously absent in both – increased their similarity[2]. Other systematic violations of subtractivity and the triangle inequality are discussed in Krantz et al. (1971, pp. 200-207).

These violations led to the development of features representations which offer an alternative approach to proximity measurement. They are neither metric nor

[2]Tversky and Krantz (1969) conducted a very similar study in which the dimensions were long vs. wide face, empty vs. filled eyes and straight vs. curved mouth. This study provided strong support for both, independence and interdimensional additivity. Krantz et al. (1971, p. 204) argue that this is due to the nature of the attributes: in the first study, the attributes were additive (e.g., presence or absence of glasses), whereas in this study, they were substitutive for a face.

dimensional in the usual sense; instead, they are set-theoretical and assess similarity by weighting common and distinctive features. These models are discussed in (Krantz et al., 1971, Chapter 14.7) and solve a lot of the violations described above.

In this thesis, we have presented metric and dimensional models for proximity measurement and provided insights into representational measurement and how such models are formalized and proved. As we have seen, geometrical models are mathematically and intuitively appealing, but they pose several difficulties in their practical usage. Finally, we would like to add, that some of these difficulties arise from the fundamental approach itself, which focuses on representing empirical structures within already existing mathematical structures. However, taking a different perspective and developing new mathematical models that fit the empirical structures could prove to be useful.

Bibliography

Aczel, J., & Dhombres, J. (1989). *Functional Equations in Several Variables*. Cambridge University Press. https://doi.org/10.1017/CBO9781139086578

Aczél, J. (1966). *Lectures on functional equations and their applications*. Academic press.

Beals, R., Krantz, D. H., & Tversky, A. (1968). Foundations of multidimensional scaling. *Psychological Review*, *75*(2), 127–142. https://doi.org/10.1037/h0025470

Breinbauer, I. (1976). Bernhard orth: Einführung in die theorie des messens. kohlhammer verlag, stuttgart/berlin/köln/mainz 1974, 131 seiten, 15, – dm (kohlhammer standards psychologie, basisbücher und studientexte). *Vierteljahrsschrift für wissenschaftliche Pädagogik*, *52*(1), 106–107. https://doi.org/10.30965/25890581-05201016

Bullen, P. S. (2003). *Handbook of Means and Their Inequalities*. Springer Netherlands. https://doi.org/10.1007/978-94-017-0399-4

Busemann, H. (1955). *The geometry of geodesics* (Vol. 6).

Campbell, N. R. (1957). Physics the Elements, 1920; reprinted as Foundations of Science. *New York*.

Cauchy, A. L. B. (1821). *Cours d'analyse de l'École Royale Polytechnique*. Imprimerie royale.

Gati, I., & Tversky, A. (1984). Weighting common and distinctive features in perceptual and conceptual judgments. *Cognitive Psychology*, *16*(3), 341–370. Retrieved March 2, 2025, from https://www.sciencedirect.com/science/article/pii/0010028584900136

Hardy, G., Littlewood, J., & Pólya, G. (1952). *Inequalities*. Cambridge University Press. https://books.google.de/books?id=t1RCSP8YKt8C

Hug, D., & Weil, W. (2020). *Lectures on Convex Geometry* (Vol. 286). Springer International Publishing. https://doi.org/10.1007/978-3-030-50180-8

Jaworska, N., & Chupetlovska-Anastasova, A. (2009). A Review of Multidimensional Scaling (MDS) and its Utility in Various Psychological Domains. *Tutorials in Quantitative Methods for Psychology*, *5*(1), 1–10. https://doi.org/10.20982/tqmp.05.1.p001

Krantz, D. H., Suppes, P., & Luce, R. D. (1971). *Foundations of Measurement: Volume 2*. Elsevier Science.

Krantz, D. H., Suppes, P., & Luce, R. D. (1989). *Foundations of measurement, Vol. 2: Geometrical, threshold, and probabilistic representations*. Academic Press.

Krantz, D. H., & Tversky, A. (1975). Similarity of rectangles: An analysis of subjective dimensions. *Journal of mathematical Psychology*, *12*(1), 4–34. Retrieved March 1, 2025, from https://www.sciencedirect.com/science/article/pii/0022249675900474

Luce, D., Krantz, D., Suppes, P., & Tversky, A. (1990). Foundations of measurement, Vol. III: Representation, axiomatization, and invariance.

Luce, R. D., & Suppes, P. (2002). Representational Measurement Theory. In *Stevens' Handbook of Experimental Psychology*. John Wiley & Sons, Ltd. https://doi.org/10.1002/0471214426.pas0401

Mulholland, H. P. (1949). On Generalizations of Minkowski's Inequality in the Form of a Triangle Inequality. *Proceedings of the London Mathematical Society*, *s2-51*(1), 294–307. https://doi.org/10.1112/plms/s2-51.4.294

Stevens, S. S. (1951). Mathematics, measurement, and psychophysics. In *Handbook of experimental psychology* (pp. 1–49). Wiley.

Stevens, S. S. (1975). Psychophysics: Introduction to its perceptual, neural, and social prospects.

Stevens, S. S. (1946). On the theory of scales of measurement. *Science*, *103*(2684), 677–680.

Thomson, B. S., Bruckner, J. B., & Bruckner, A. M. (2008). *Elementary real analysis* (Vol. 1). ClassicalRealAnalysis. com.

Tversky, A. (1977). Features of similarity. *Psychological review*, *84*(4), 327.

Tversky, A., & Gati, I. (1979). Studies of similarity. In *Cognition and categorization* (pp. 79–98). Routledge. Retrieved March 2, 2025, from https://www.taylorfrancis.com/chapters/edit/10.4324/9781032633275-7/studies-similarity-amos-tversky-itamar-gati

Tversky, A., & Krantz, D. H. (1969). Similarity of schematic faces: A test of. *Perception & Psychophysics*, *5*(2), 124–128. https://doi.org/10.3758/BF03210535

Tversky, A., & Krantz, D. H. (1970). The dimensional representation and the metric structure of similarity data. *Journal of mathematical psychology*, *7*(3), 572–596.

Wender, K. (1971). A test of independence of dimensions in multidimensional scaling. *Perception & Psychophysics*, *10*(1), 30–32. https://doi.org/10.3758/BF03205762

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Erklärung zur Abschlussarbeit gemäß § 22 Abs. 7 APB TU Darmstadt

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Bei einer Thesis des Fachbereichs Architektur entspricht die eingereichte elektronische Fassung dem vorgestellten Modell und den vorgelegten Plänen.

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For a thesis of the Department of Architecture, the submitted electronic version corresponds to the presented model and the submitted architectural plans.

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